



INDUSTRIAL INTERNSHIP

HINDUSTAN PETROLEUM CORPORATION
LIMITED -VISAKHAPATNAM



Industrial Internship

RESIDUE UPGRADATION TECHNOLOGIES

SUBMITTED BY

MENAVATH PAVAN KUMAR(230008021)

DEPARTMENT : TECHNICAL

DURATION : 2 Months(25-05-2026 to 25-07-2026)

DEGREE & BRANCH : B.Tech & CHEMICAL ENGINEERING

COLLAGE : INDIAN INSTITUTE OF TECHNOLOGY INDORE

SUPERVISED BY:

MR.SUDHIR ANAMANAMURI

MANAGER-PROCESS ANALYSIS AND DESIGN

HPCL-VR

CERTIFICATE

This is to certify that Mr.Menavath Pavan kumar, a student of Bachelor of Technology in Chemical Engineering from Indian institute of Technology Indore, has successfully completed an internship at Hindustan Petroleum Corporation limited Visakhapatnam(HPCL-VR) from 25-05-2026 to 25-07-2026. During the internship, he worked on “Residue Upgradation Technologies ” and performed the assigned tasks.

Overall Performance Grade: [___ / 10]

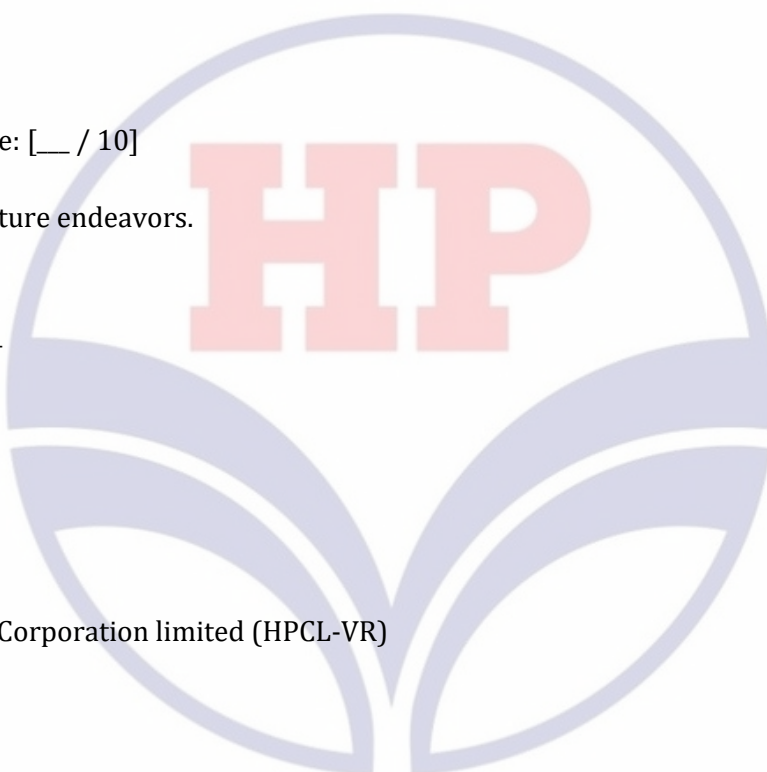
We wish him success in future endeavors.

Signature: _____

Name: _____

Designation: _____

For Hindustan Petroleum Corporation limited (HPCL-VR)



DECLARATION

I, MENAVATH PAVAN KUMAR (Roll No: 230008021), student of Bachelor of Technology in Chemical Engineering from Indian Institute of technology, hereby declare that the industrial internship report entitled “RESIDUE UPGRADATION TECHNOLOGIES IN PETROLEUM REFINERIES WITH SPECIAL REFERENCE TO HPCL VISAKH REFINERY”, submitted to the Department of Chemical Engineering, is an authentic record of the study and technical understanding developed by me during my industrial internship at Hindustan Petroleum Corporation Limited (HPCL), Visakh Refinery.

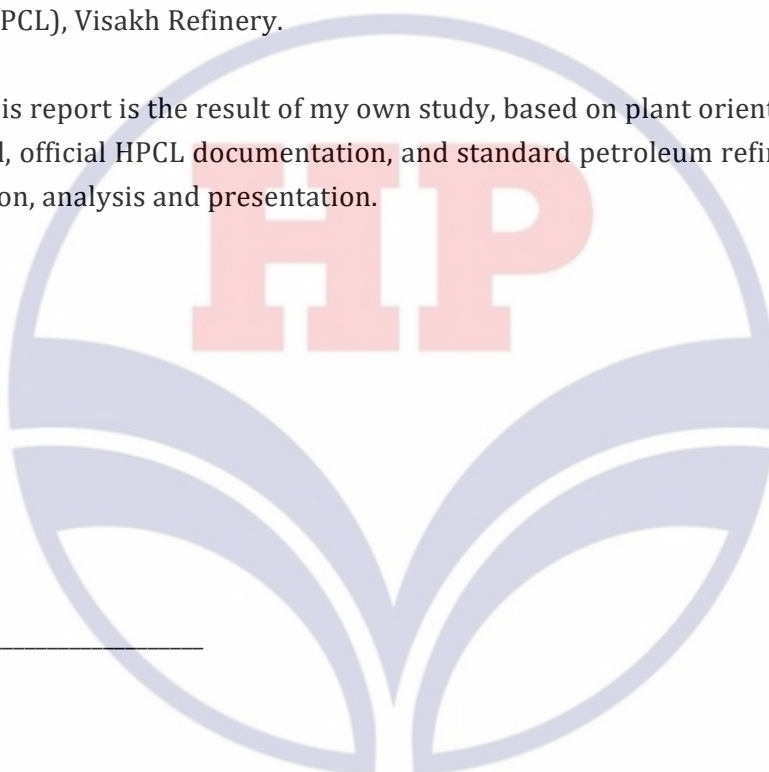
I further declare that this report is the result of my own study, based on plant orientation, technical discussions with refinery personnel, official HPCL documentation, and standard petroleum refining literature. This work is original in its compilation, analysis and presentation.

Place: Visakhapatnam

Date: 25/07/2026

Student signature: _____

Roll No: 230008021



ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to Hindustan Petroleum Corporation Limited (HPCL), Visakh Refinery, for granting me the opportunity to undergo my industrial internship in a plant environment of such technical eminence, and for exposing me to the operational and technological depth of one of India's most advanced deep-conversion refineries.

I am deeply thankful to my mentor Sudhir Anamanamuri, Manager – Process Analysis and Design, for his valuable guidance, patient clarification of process concepts, and constant encouragement throughout the internship period. His insights into the Residue Upgradation Facility (RUF) and the LC-Max™ Resid Hydrocracking Unit were instrumental in shaping the technical direction of this report.

I would also like to place on record my appreciation for the officers and field crew of the Crude Distillation Unit (CDU), Vacuum Distillation Unit (VDU), Full Conversion Hydrocracker Unit (FCHCU) and Residue Upgradation Facility (RUF) for sharing their operational knowledge and for patiently answering my queries regarding process philosophy, equipment design and safety practices.

I sincerely thank the Human Resources (HR) Department of HPCL Visakh Refinery for the concern shown in granting me this internship opportunity and for the administrative support extended throughout my tenure.

Finally, I would like to thank the faculty members of the Department of Chemical Engineering, Indian Institute of Technology Indore.

Menavath Pavan Kumar

ABSTRACT

The petroleum refining industry is progressively shifting from a fuel-oil-driven business model towards a distillate- maximization and petrochemical-integration model, driven by declining global demand for high-sulphur fuel oil, tightening environmental legislation (IMO 2020, BS-VI), and the economic imperative to extract maximum value from every barrel of crude oil processed. Central to this transition is the technology of residue upgradation – the conversion of the heaviest, lowest-value fraction of the barrel (atmospheric and vacuum residue) into high-value transportation fuels, petrochemical feedstocks and clean fuel components.

This report presents a comprehensive technical study of residue upgradation technologies employed in modern petroleum refineries, undertaken during an industrial internship at Hindustan Petroleum Corporation Limited (HPCL), Visakh Refinery. The report begins with an overview of HPCL and the historical evolution of the Visakh Refinery, including the Visakh Refinery Modernization Project (VRMP), which enhanced the refinery's crude processing capacity to 15 MMTPA and its Nelson Complexity Index to 11.6.

The scientific and engineering fundamentals of petroleum residues – their composition, Conradson Carbon Residue (CCR), metals content, asphaltene and resin fractions – are examined in detail, followed by an evaluation of the principal residue conversion routes: thermal processes (visbreaking, delayed coking, fluid coking, flexicoking), catalytic process (Residue Fluid Catalytic Cracking), hydrogen addition process (ebullated-bed resid hydrocracking technologies such as LC-FINING, H-Oil and LC-MAX™), and solvent-based deasphalting processes.

A dedicated chapter is devoted to HPCL Visakh Refinery's own Residue Upgradation Facility (RUF), commissioned in January 2026, which houses India's first Resid Hydrocracking Unit and the world's first LC-Max™ unit, achieving approximately 93% conversion of bottom-of-the-barrel oils into high-value distillates. The report discusses the reaction chemistry (cracking, hydrogenation, hydrodesulphurization, hydrodenitrogenation, hydrodemetallization and coke-formation mechanisms), representative process calculations, environmental and sustainability aspects, industrial safety practices associated with high-pressure hydroprocessing, and the future outlook for residue upgradation technology, including digitalization and low-carbon refining pathways.

The study concludes that residue upgradation, and in particular hydrogen-based ebullated-bed hydrocracking technology such as LC-Max™, represents the most effective route for Indian refiners to maximize distillate yield, comply with BS-VI clean-fuel norms, and remain economically and environmentally competitive in a carbon- constrained future.

CONTENTS

DECLARATION-----	03
ACKNOWLEDGEMENT-----	04
ABSTRACT-----	04
LIST OF FLOW CHARTS-----	06
LIST OF TABLES-----	08
ABBREVIATIONS-----	09
 CHAPTER 1: INTRODUCTION TO HPCL AND VISAKH REFINERY-----	13
1.1 HPCL Corporate Overview - 1.6 Refinery Complexity	
 CHAPTER 2: PETROLEUM RESIDUES - COMPOSITION AND CHARACTERIZATION-----	16
2.1 Definition-2.5 Metals Content	
 CHAPTER 3: NEED FOR RESIDUE UPGRADATION-----	19
3.1 Global Trends - 3.6 Summary of Drivers	
 CHAPTER 4: RESIDUE UPGRADATION TECHNOLOGIES-----	21
4.1 Thermal Processes - 4.6 Comparative Assessment	
 CHAPTER 5: REACTION CHEMISTRY OF RESIDUE UPGRADATION-----	32
5.1 Thermal Cracking - 5.7 Coke Formation & Catalyst Deactivation	
 CHAPTER 6: HPCL VISAKH REFINERY AND RESIDUE UPGRADATION-----	35
6.1 RUF Overview -6.6 Strategic Significance	
 CHAPTER 7: PROCESS CALCULATIONS-----	38
7.1 API Gravity-7.7 Economics	
 CHAPTER 8: ENVIRONMENTAL AND SUSTAINABILITY ASPECTS-----	41
 CHAPTER 9: INDUSTRIAL SAFETY IN RESIDUE UPGRADATION OPERATIONS-----	43
 CHAPTER 10: FUTURE SCOPE-----	44
 CHAPTER 11: CONCLUSION-----	45
 CHAPTER 12:REFERENCES-----	46

LIST OF FLOW CHARTS

Chart: No.	Caption	Page
1	Expansion of Visakh Refinery (1957–2026)	13
2	Post-VRMP Block Flow Diagram, HPCL Visakh Refinery	14
3	Composition Envelope of Atmospheric and Vacuum Residue	16
4	SARA Fraction Structure (Saturates, Aromatics, Resins, Asphaltenes)	18
5	Global Bottom-of-the-Barrel Conversion Routes – Classification	22
6	Visbreaking Unit – Simplified Process Flow Diagram	23
7	Delayed Coking Unit – Simplified Process Flow Diagram	24
8	Fluid Coking / Flexicoking – Simplified Process Flow Diagram	25
9	Residue Fluid Catalytic Cracking (RFCC) – Simplified Process Flow Diagram	26
10	Ebullated-Bed Resid Hydrocracking – Generic Reactor Concept	27
11	LC-MAX™ / LC-FINING™ Ebullated-Bed Process Flow Diagram	28
12	Solvent Deasphalting Unit – Simplified Process Flow Diagram	29
13	Reaction Pathway – Catalytic Hydrocracking of Residue Molecules	33

15	Catalyst Deactivation Mechanisms in Resid Hydroprocessing	34
16	HPCL Visakh Refinery – Residue Upgradation Facility (RUF) Block Diagram	37
17	LC-MAX™ Reactor Train at HPCL Visakh Refinery – Conceptual Layout	37
18	Product Slate Shift – Pre-VRMP vs Post-RUF Visakh Refinery	38
19	Mass Balance Envelope for a Typical Resid Hydrocracking Unit	41
20	Refinery Complexity Growth – Nelson Complexity Index Trend, Visakh Refinery	43

LIST OF TABLES

Table No.	Title	Page
1	HPCL Visakh Refinery – Crude Distillation Capacity by Unit	14
2	HPCL Visakh Refinery – Secondary Processing Unit Capacities	15
3	Typical Properties of Atmospheric Residue (AR) and Vacuum Residue (VR)	17
4	Classification of Crude Oil / Residue by CCR and Metals Content	19
5	SARA Fraction Ranges in Typical Vacuum Residues	18
6	Comparison of Thermal Residue-Conversion Processes	23
7	Typical Yield Distribution – Delayed Coking of Vacuum Residue	24
8	Comparison of Catalytic and Hydrogen-Addition Residue Upgradation Routes	31
9	Typical Operating Conditions – Ebullated-Bed Resid Hydrocracking (LC-MAX™ type)	28
10	Comparative Conversion and Product Quality – SDA, RFCC, Coking, Resid Hydrocracking	32
11	HPCL Visakh Refinery – Residue Upgradation Facility (RUF) Design Data	36
12	Sample CCR and API Gravity Calculation Data Set	39
13	Sample Hydrogen Consumption Estimate for Resid Hydrocracking	40

ABBREVIATIONS

Abbreviation	Full Form
AR	Atmospheric Residue
VR / VB	Vacuum Residue / Vacuum Bottoms
VGO	Vacuum Gas Oil
LVGO	Light Vacuum Gas Oil
HVGO	Heavy Vacuum Gas Oil
CDU	Crude Distillation Unit
VDU	Vacuum Distillation Unit
FCC	Fluid Catalytic Cracking
RFCC	Residue Fluid Catalytic Cracking
HCU	Hydrocracking Unit
FCHCU	Full Conversion Hydrocracking Unit
DCU	Delayed Coking Unit
SDA	Solvent Deasphalting
RUF	Residue Upgradation Facility

LC-FINING™	Chevron Lummus Global Ebullated-Bed Resid Hydrocracking Technology
LC-MAX™	Chevron Lummus Global High-Conversion Ebullated-Bed Resid Hydrocracking Technology
H-Oil®	Axens/CLG Ebullated-Bed Resid Hydrocracking Technology
CLG	Chevron Lummus Global (Licensor)
HGU	Hydrogen Generation Unit
DHDT / DHT	Diesel Hydrotreating Unit
DHDS	Diesel Hydro-Desulphurization Unit
SRU	Sulphur Recovery Unit
PRU	Propylene Recovery Unit
BBU	Bitumen Blowing Unit
VRMP	Visakh Refinery Modernization Project
CCR	Conradson Carbon Residue
MCR	Micro Carbon Residue
SARA	Saturates, Aromatics, Resins, Asphaltenes

HDS	Hydrodesulphurization
HDN	Hydrodenitrogenation
HDM	Hydrodemetallization
HDA	Hydrodearomatization
BS-VI	Bharat Stage VI Emission Norms
MMTPA	Million Metric Tonnes Per Annum
TMT / TPD / TPA	Thousand Metric Tonnes / Tonnes Per Day / Tonnes Per Annum
NCI	Nelson Complexity Index
GRM	Gross Refining Margin
EII	Energy Intensity Index
WHSV / LHSV	Weight / Liquid Hourly Space Velocity
ppm(w)	Parts Per Million by Weight
IMO	International Maritime Organization
VLSFO	Very Low Sulphur Fuel Oil
LSHS	Low Sulphur Heavy Stock

API	American Petroleum Institute (Gravity Scale)
EOR / TAN	End-of-Run / Total Acid Number
PSA	Pressure Swing Adsorption



CHAPTER 1: INTRODUCTION TO HPCL AND VISAKH REFINERY

1.1 Hindustan Petroleum Corporation Limited – Corporate Overview

Hindustan Petroleum Corporation Limited (HPCL) is a Maharatna Central Public Sector Enterprise under the Ministry of Petroleum and Natural Gas, Government of India, and is the second-largest integrated oil refining and marketing company in the country. HPCL's business segments span crude oil refining, marketing of petroleum products (retail, industrial, aviation and lubricants), LPG distribution, cross-country pipeline transportation, and an expanding portfolio of petrochemicals, city gas distribution and renewable energy. HPCL holds close to 20% of India's refining market share and operates an extensive marketing infrastructure comprising retail outlets, LPG bottling plants, aviation fuel stations, lube blending plants and cross-country pipelines.

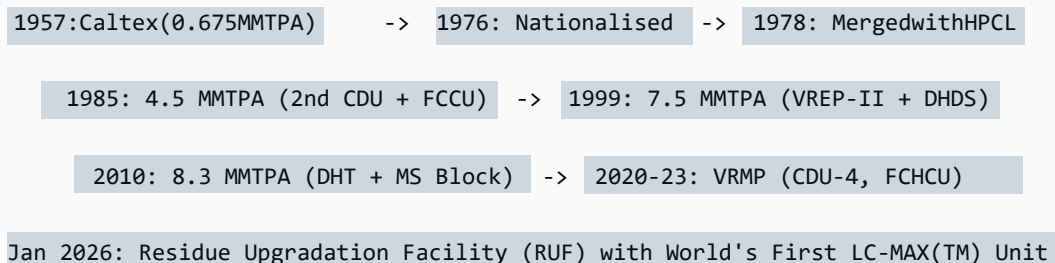
HPCL operates two major refineries – the Mumbai Refinery (7.5 MMTPA) on the west coast, and the Visakh Refinery (15 MMTPA post-VRMP) on the east coast – in addition to joint-venture refineries at Bathinda (HMEL) and Barmer (HPCL-Rajasthan Refinery). Its Mumbai lube refinery, with a capacity of about 3,33,500 TPA, is the largest lube-base-oil refinery in the country.

1.2 Visakh Refinery – Historical Evolution

Visakh Refinery, located on the east coast of India at Visakhapatnam, Andhra Pradesh, is the oldest refinery on the eastern seaboard and the first major industrial establishment in the city. It was originally commissioned in 1957 as Caltex Oil Refining (India) Limited (CORIL) with an installed capacity of 0.675 MMTPA. Following nationalization of Caltex's Indian operations in 1976, the refinery was amalgamated with HPCL in 1978.

The refinery has undergone successive phases of capacity expansion and technological upgradation: 1.5 MMTPA through debottlenecking in the years following nationalization, 4.5 MMTPA in 1985 with the commissioning of a second crude train and an FCC Unit, 7.5 MMTPA in 1999 under the Visakh Refinery Expansion Project (VREP-II) together with a Diesel Hydro-Desulphurization (DHDS) unit, and 8.3 MMTPA by 2010 with the addition of Diesel Hydrotreating (DHT) facilities and the associated MS-quality upgradation block.

Flow-Chart 1: Expansion of Visakh Refinery (1957–2026)



Source: HP CL Visakh Refinery corporate presentation; HP CL press release, 05 January 2026.

1.3 Visakh Refinery Modernization Project (VRMP)

The Visakh Refinery Modernization Project (VRMP) is a landmark expansion-cum-modernization initiative undertaken to increase the refinery's crude processing capacity from 8.3 MMTPA to 15 MMTPA, at a reported

project cost of approximately ₹26,264 crore. The stated objectives of VRMP were: compliance with Bharat Stage VI (BS-VI) fuel specifications; increase of refining capacity to 15 MMTPA; improvement of distillate yield from about 76 wt% to 86 wt% on crude, with associated bottoms upgradation; maintenance of SO_x emissions at the existing permissible ceiling of 11.5 tonnes per day; and improvement of refinery margins from roughly US\$6.8/bbl to US\$13.72/bbl.

VRMP introduced a new 9 MMTPA fourth Crude Distillation Unit (CDU-IV, commissioned March 2023, licensed by EIL and built by L&T), a 3.05–3.06 MMTPA Full Conversion Hydrocracking Unit (FCHCU, commissioned October 2023, UOP technology, L&T as EPC contractor), a Naphtha Isomerization Unit (NIU), revamped Diesel Hydrotreating (DHDT), Naphtha Hydrotreating/CCR Platforming (NHT/CCR) and FCC-NHT blocks, augmented Hydrogen Generation and Sulphur Recovery capacity, and – as the capstone bottoms-upgradation investment – the Residue Upgradation Facility (RUF), which was commissioned in January 2026 around the world's first LC-MAX™ Resid Hydrocracking Unit. India's tallest CDU column (75.51 m length, 7 m diameter, 685 MT) and India's widest vacuum column (VDU: 62.52 m length, 12 m diameter, 923 MT) were engineered and erected as single-piece modules as part of VRMP, reflecting the scale of the project.

1.4 Present Refinery Configuration and Capacities

Post-VRMP and post-RUF commissioning, HPCL Visakh Refinery operates as a highly complex, deep-conversion refinery with an aggregate crude distillation capacity of approximately 15 MMTPA distributed across three parallel crude trains, together with an integrated secondary processing complex covering catalytic cracking, hydrocracking, resid hydrocracking, hydrotreating, reforming and sulphur/hydrogen utilities.

Table 1: HPCL Visakh Refinery – Crude Distillation Capacity by Unit

Unit	Nominal Capacity (MMTPA)	Remarks
CDU-2	3.0	Atmospheric + Vacuum distillation train
CDU-3	3.0	Atmospheric + Vacuum distillation train
CDU-4 (VDU-4)	9.0	Commissioned March 2023 under VRMP; India's tallest CDU column
Total	15.0	Post-VRMP aggregate crude processing capacity

Source: HPCL Visakh Refinery presentation notes; VRMP disclosures.

Table 2: HPCL Visakh Refinery – Secondary Processing Unit Capacities (Indicative)

Unit	Capacity	Licensor
FCCU-I	1.0 MMTPA	ExxonMobil
FCCU-II	0.96-0.97 MMTPA	UOP
FCHCU (Full Conversion Hydrocracker)	~3.05-3.06 MMTPA	UOP
Residue Upgradation Facility – LC-MAX™ Resid Hydrocracker (RUF)	3.55 MMTPA	CLG (Chevron Lummus Global)
Visbreaker Unit (VBU)	1.0 MMTPA	-
Naphtha Isomerization Unit (NIU)	0.35 MMTPA	UOP
DHDT (Diesel Hydrotreating, revamped)	2.86 MMTPA	UOP
DHDS (Diesel Hydro-Desulphurization)	2.4 MMTPA	-
NHT/CCR Platformer (revamped)	1.5 MMTPA	UOP
FCC-NHT (revamped)	0.893 MMTPA	-
Bitumen Blowing Unit (BBU)	225 TMTA	Axens
Propylene Recovery Unit (PRU)	23,000-65,000 TPA	Polymer
Hydrogen Generation Unit (HGU-2)	2 × 113 KTPA	- Technip Benelux

Source: HPCL Visakh Refinery corporate presentation notes; VRMP process-unit disclosures; HPCL press release, January 2026.

1.5 Products and BS-VI Fuel Capability

The product slate of Visakh Refinery spans the entire barrel: light ends (LPG, propylene, naphtha, motor spirit), middle distillates (superior kerosene oil, aviation turbine fuel, high-speed diesel, light diesel oil, mineral turpentine oil, jute batching oil) and heavy ends (fuel oil, low-sulphur heavy stock, very-low-sulphur fuel oil, bitumen and elemental sulphur). Following the commissioning of DHDS, DHT and the NHT/CCR/Isom block under VRMP, the refinery is fully equipped to produce BS-VI-compliant motor spirit (MS) and high-speed diesel (HSD), with sulphur specification limited to 10 ppm(w). The commissioning of the RUF has further shifted the refinery's yield structure towards middle distillates, reducing the proportion of low-value fueloil inthe product basket.

1.6 Refinery Complexity and Strategic Position

With thecommissioning of the RUF, HPCL Visakh Refinery's Nelson Complexity Index (NCI) – a widely used industry metric that weights each secondary processing unit according to its cost and conversion intensity relative to primary distillation – has risen to 11.6, placing the refinery among the most complex, deep-conversion refineries in India. A higher NCI reflects greater flexibility to process heavier and more economically discounted (“opportunity”) crude grades while still producing a light, clean, high-value product slate, which is the central theme explored in this report.

CHAPTER 2: PETROLEUM RESIDUES – COMPOSITION AND CHARACTERIZATION

2.1 Definition of Refinery Residues

A petroleum “residue” is the non-volatile, high-boiling fraction remaining after crude oil has been subjected to distillation. Two residues are relevant to refinery bottoms processing: Atmospheric Residue (AR), also called “long residue”, which is the bottom product of the atmospheric (crude) distillation column, comprising material boiling above approximately 350–370°C; and Vacuum Residue (VR), also called “short residue”, which is the bottom product of the vacuum distillation column operating on AR feed, comprising material boiling above approximately 520– 565°C (true boiling point, atmospheric-equivalent). VR typically constitutes 15–35 wt% of crude oil depending on crudetype, andisthe principalfeedstocktomodern residue upgradation units.

2.2 Physical and Chemical Properties

Vacuum residue is characterized by very high viscosity, high density (low API gravity, typically 5–15°API), high Conradson Carbon Residue (CCR), elevated concentrations of sulphur, nitrogen and metallic contaminants (principally nickel and vanadium), and a high content of large, polynuclear-aromatic, heteroatom-substituted molecules collectively described through the SARA fractionation – Saturates, Aromatics, Resins and Asphaltenes.

Flow Chart 3: Composition Envelope of Atmospheric and Vacuum Residue

CRUDE OIL (100%)

-> Atmospheric Distillates (LPG, Naphtha, Kero, Diesel)	~45-65 wt%
-> Atmospheric Residue (AR, 350C+)	~35-55 wt%
-> Vacuum Distillates (LVGO, HVGO)	~15-25 wt%
-> Vacuum Residue (VR, 520-565C+)	~15-35 wt% <- RUF FEED

Source: Generalized composition envelope; Gary, Handwerk & Kaiser, Petroleum Refining: Technology and Economics.

Table 3: Typical Properties of Atmospheric Residue (AR) and Vacuum Residue (VR)

Property	Atmospheric Residue (AR)	Vacuum Residue (VR)
API Gravity	12 – 22°	2 – 15°
Density @15°C	0.92 – 0.98 g/cc	0.98 – 1.05 g/cc
Conradson Carbon Residue (CCR)	8 – 14 wt%	15 – 30 wt%
Sulphur	2 – 5 wt%	3 – 6 wt%
Nitrogen (basic + total)	0.2 – 0.5 wt%	0.4 – 0.9 wt%
Nickel + Vanadium	20 – 100 ppm	80 – 400+ ppm
Asphaltenes (n-C7 insolubles)	2 – 6 wt%	6 – 18 wt%
Viscosity @100°C	50 – 400 cSt	500 – 5000+ cSt
Pour Point	20 – 45°C	30 – 60°C

Source: Representative industry values compiled from standard refining literature (Speight, The Chemistry and Technology of Petroleum; Gary & Handwerk).

2.3 SARA Fractionation

The SARA analysis (ASTM D2007 / IP 143 based methods) separates residue into four fractions of increasing polarity and molecular complexity:

- Saturates: paraffinic and naphthenic hydrocarbons, low polarity, good hydrogen-donor character during hydroprocessing.
- Aromatics: mono- and polynuclear aromatic ring systems; a significant proportion is converted to lighter products but also contributes to coke precursors at high aromaticity.
- Resins: polar, moderately high molecular weight molecules containing heteroatoms (S, N, O) that act as peptizing (stabilizing) agents for asphaltenes in the oil matrix.
- Asphaltenes: the highest molecular weight, most polar, most condensed polynuclear-aromatic fraction, insoluble in n-heptane/n-pentane; the principal precursor of coke and catalyst-fouling deposits during thermal and catalytic processing.

Flow Chart 4: SARA Fraction Structure (Saturates, Aromatics, Resins, Asphaltenes)

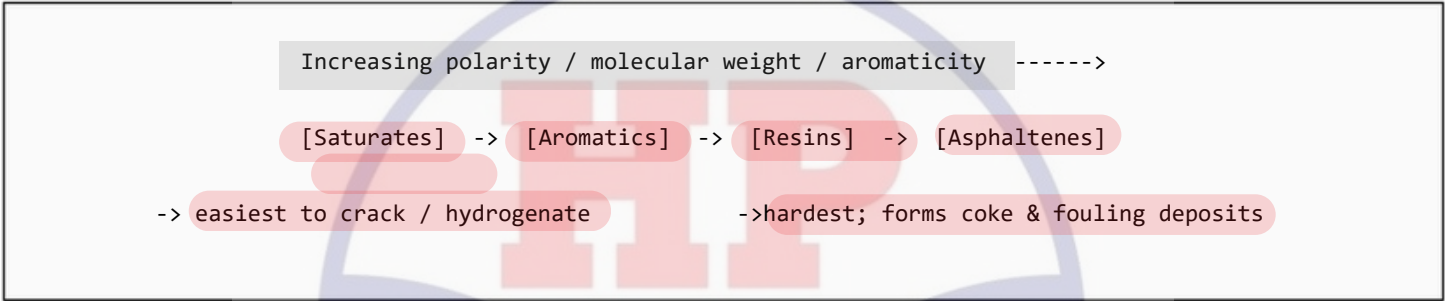


Table 5: SARA Fraction Ranges in Typical Vacuum Residues

Fraction	Typical Range (wt% of VR)
Saturates	10 – 25%
Aromatics	35 – 55%
Resins	15 – 30%
Asphaltenes	6 – 18%

2.4 Conradson Carbon Residue (CCR) and its Significance

Conradson Carbon Residue is measured (ASTM D189/D4530) by pyrolyzing a weighed sample of oil in the absence of air; the mass of carbonaceous residue remaining, expressed as a percentage of the original sample, is a strong

empirical predictor of coke yield in downstream thermal and catalytic units. High-CCR residues (VR with CCR > 20 wt%) are generally routed to thermal or hydrogen-addition conversion units, since fluid catalytic cracking of such feeds leads to excessive catalyst-coke deposition and unacceptable regenerator heat balance.

2.5 Metals Content and Poisoning Potential

Nickel and vanadium, present in residue predominantly as metalloporphyrin complexes, are of particular concern because they deposit irreversibly on catalyst surfaces during hydroprocessing and catalytic cracking, blocking active sites and pore mouths (metal poisoning) and, in the case of vanadium, promoting dehydrogenation side reactions that increase hydrogen and coke yields. Metals content is therefore a primary basis for selecting between disposable- catalyst thermal routes, catalyst-replenishing ebullated-bed hydrocracking, and metal-rejecting solvent deasphalting

Table 4: Classification of Crude Oil / Residue by CCR and Metals Content

Category	CCR (wt%)	Ni+V (ppm)	Preferred Upgradation Route
Light / sweet residue	< 10	< 50	RFCC, mild hydrotreating
Medium residue	10 – 20	50 – 150	RFCC (with feed pretreat), visbreaking, coking
Heavy / high-metals residue	20 – 30	150 – 300	Delayed coking, ebullated-bed hydrocracking (LC-MAX™/H-Oil), SDA
Extra-heavy residue	> 30	> 300	Coking, SDA + coking, gasification

CHAPTER 3: NEED FOR RESIDUE UPGRADATION

3.1 Global Refining Trends and Bottom-of-the-Barrel Conversion

Global crude oil quality has been gradually heavying, with average API gravity declining and sulphur content rising as easily accessible light-sweet reserves deplete and refiners increasingly process discounted “opportunity” crudes and blends. At the same time, global demand for high-sulphur fuel oil has structurally declined, accelerated by the International Maritime Organization's IMO 2020 regulation capping marine fuel sulphur at 0.5 wt% (0.1 wt% in Emission Control Areas), which effectively stranded a large volume of historical high-sulphur fuel oil (HSFO) demand. Refiners that lack bottom-of-the-barrel conversion capacity are structurally exposed to this shrinking, low-value fuel- oil market, whereas refiners with residue upgradation capacity can convert the same barrel of crude into high-value diesel, gasoline and petrochemical feedstock.

3.2 Economic Importance – Refinery Margin Improvement

The economic rationale for residue upgradation rests on the substantial value differential between vacuum residue / fuel oil and light-to-middle distillates. In most refining margin structures, fuel oil realizes a discount of US\$10–25/bbl relative to crude, while diesel and gasoline typically realize a premium. Converting a barrel of VR (worth close to, or below, crude price) into diesel and naphtha (worth substantially more) directly captures this margin uplift. HPCL has explicitly stated that its RUF investment at Visakh Refinery is expected to substantially increase Gross Refining Margins (GRM) through a superior product slate, flexibility to process heavier/opportunity crudes, and higher value realization per barrel.

3.3 Fuel Demand Shift Towards Middle Distillates

India's transportation-fuel demand is dominated by diesel (HSD) and, to a lesser but growing extent, aviation turbine fuel and petrochemical naphtha, while demand for heavy fuel oil in domestic industry has been steadily eroded by grid electrification, natural gas substitution and environmental regulation. Refiners are therefore incentivized to maximize distillate yield per barrel of crude – a strategy explicitly described by HPCL as “diesel maximization” in the context of its Visakh Refinery bottoms-upgradation investments.

3.4 Clean Fuel Regulations – BS-VI Norms

The adoption of Bharat Stage VI (BS-VI) emission norms across India from April 2020 mandated a step reduction in sulphur content of motor spirit and diesel to a maximum of 10 ppm(w), alongside tighter aromatics and cetane/octane specifications. Meeting BS-VI specifications requires refinery-wide integration of deep hydrotreating (DHDS/DHT), isomerization/reforming for octane recovery, and – critically – a residue upgradation scheme that does not merely convert VR to fuel oil but produces intermediate streams (unconverted oil, resid- derived VGO, diesel) of a quality suitable for downstream hydrotreating to BS-VI specification. Ebullated-bed resid hydrocracking technologies such as LC-MAX™ are particularly well suited to this requirement because, in addition to converting VR, they substantially hydrotreat the unconverted oil and gas-oil fractions in the same reactor train, reducing the downstream hydrotreating severity required.

3.5 Refinery Margin and Competitiveness – Complexity as a Strategic Lever

Refineries are frequently benchmarked using the Nelson Complexity Index (NCI), a weighted measure of secondary conversion capacity relative to primary distillation capacity. Complex, deep-conversion refineries consistently outperform simple “hydroskimming” refineries on gross refining margin because they can process a wider slate of (often more discounted) crude grades and produce a lighter, more valuable product mix. HPCL's own disclosure that the Visakh Refinery's NCI rose to 11.6 following RUF commissioning is a direct illustration of this strategic logic – positioning the refinery among India's most complex, and hence most margin-resilient, refining assets.



Table showing directional relationship between conversion investment and refining margin (illustrative, not refinery-specific figures)

Refinery Configuration	Indicative NCI	Typical Margin Character
Topping / hydroskimming refinery	2 – 4	Low, highly crude-price sensitive; limited crude flexibility
FCC-based conversion refinery	6 – 9	Moderate; good gasoline/LPG yield, residual fuel-oil exposure remains
Deep-conversion refinery with resid hydrocracking (e.g., Visakh Refinery post-RUF)	10 – 12+	High; maximized distillate yield, wide crude flexibility, reduced fuel-oil exposure

Source: Illustrative classification based on standard refining-economics literature (Jones & Pujadó, Handbook of Petroleum Processing; HPCL press release, Jan 2026).

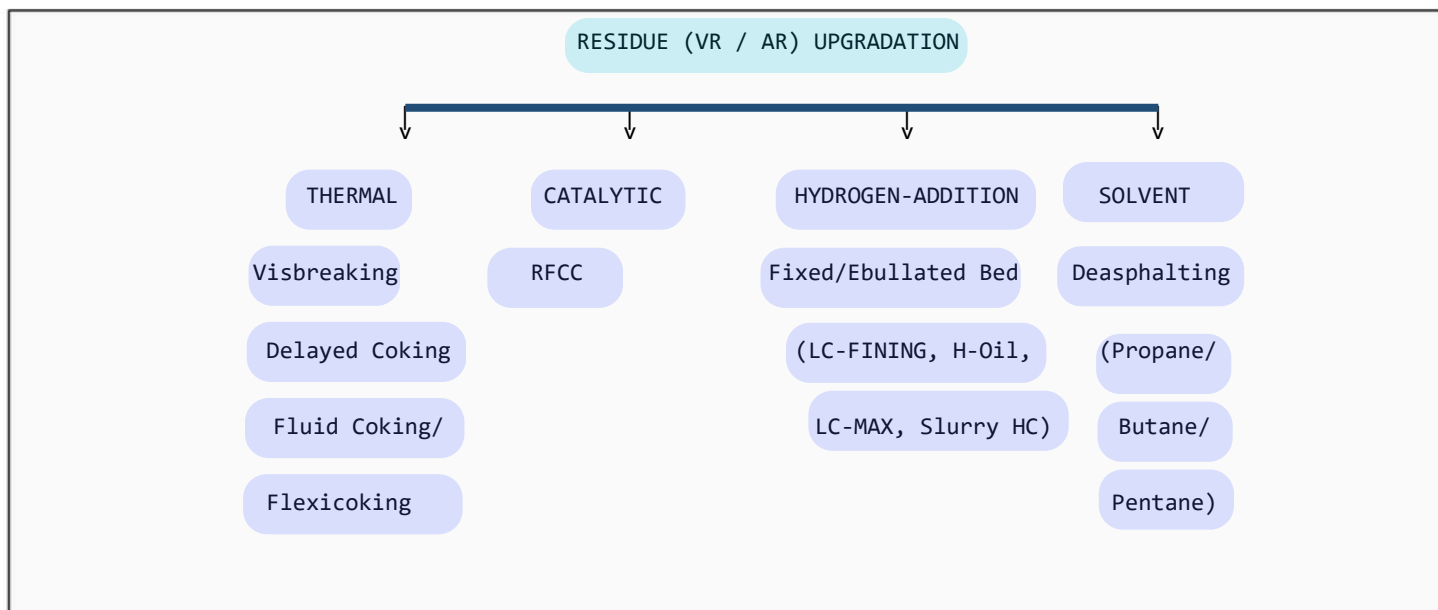
3.6 Summary of Drivers

- Structural decline in high-sulphur fuel oil demand (IMO 2020 and domestic industrial fuel substitution).
- Growing middle-distillate (diesel/ATF) demand requiring diesel-maximization refinery configurations.
- Mandatory BS-VI clean-fuel specifications requiring integrated deep hydroprocessing.
- Direct refining-margin uplift from converting low-value VR into high-value distillates.
- Strategic crude-sourcing flexibility to process heavier, discounted (“opportunity”) crude grades.
- Reduction of refinery carbon and sulphur emission intensity per barrel of finished product.

CHAPTER 4: RESIDUE UPGRADATION TECHNOLOGIES

Residue upgradation technologies are broadly classified into four families based on the mechanism by which conversion is achieved: (i) thermal processes, which rely purely on heat to crack large molecules, rejecting excess carbon as solid coke; (ii) catalytic processes, which use an acidic catalyst to promote carbocation-mediated cracking, again rejecting carbon as coke on the catalyst; (iii) hydrogen-addition processes, which supply external hydrogen under catalytic conditions to saturate cracked fragments and suppress coke formation, thereby achieving much higher liquid yields; and (iv) solvent-based processes, which achieve “conversion” not by molecular cracking but by physical rejection of the most refractory (asphaltenic) fraction into a separate, low-value pitch stream.





4.1 Thermal Processes

Thermal processes crack large residue molecules purely by the application of heat (typically 420–510°C) with sufficient residence time to achieve the desired conversion, without a catalyst. Because no hydrogen is added, the hydrogen-to-carbon ratio of the feed is redistributed by carbon rejection – lighter products become more paraffinic/hydrogen-rich while a carbon-rich, hydrogen-deficient coke or heavy tar by-product is formed.

4.1.1 Visbreaking

Working Principle: Visbreaking (“viscosity breaking”) is the mildest thermal conversion process, designed primarily to reduce the viscosity and pour point of atmospheric or vacuum residue – thereby reducing the quantity of expensive cutter stock (diesel/kerosene) required to blend the residue down to marketable fuel-oil viscosity – rather than to achieve deep molecular conversion.

Process Chemistry: Long paraffinic side-chains attached to naphthenic and aromatic ring systems undergo free-radical thermal cracking, following classical hydrocarbon pyrolysis kinetics: chain initiation by C–C bond homolysis, propagation via hydrogen abstraction and β -scission, and termination by radical recombination or disproportionation. The dominant products are light gas, naphtha, a gas-oil range fraction, and a visbroken residue (tar) of markedly reduced viscosity.

Two configurations are used industrially: coil (furnace) visbreaking, in which cracking occurs in the radiant coil of a fired heater followed by rapid quench, and soaker visbreaking, in which the feed is heated to a lower coil-outlet temperature and then held in an insulated soaker drum for extended residence time, achieving comparable conversion at lower temperature and hence lower energy consumption and fuel-gas make.



Table 6: Comparison of Thermal Residue-Conversion Processes

Parameter	Visbreaking	Delayed Coking	Fluid Coking / Flexicoking
Typical COT / operating temp.	440 – 490°C	480 – 505°C (coke drum ~415–450°C)	510 – 540°C (reactor)
Residence time	1 – 3 min (coil); hrs (soaker drum)	18 – 24 hr coking cycle per drum	Seconds (fluidized bed)
Conversion (VR basis)	10 – 20 wt%	65 – 85 wt%	85 – 95+ wt% (Flexicoking gasifies coke)
Coke / solid by-product	None (tar product)	Green petroleum coke (25–35 wt% of VR)	Coke gasified to fuel gas (Flexicoking)
Product quality	Low; olefinic, unstable	Moderate; olefinic naphtha/gas oil needs hydrotreating	Similar to delayed coking
Capital intensity	Low	High	Very high

Flow Chart 6: Visbreaking Unit – Simplified Process Flow Diagram

VR Feed -> Preheat -> Fired Heater (Coil crack, ~450–480°C) -> Soaker Drum (optional)
-> Quench -> Fractionator -> [Gas, Naphtha, Gas Oil, Visbroken Residue/Tar]

Advantages: Low capital cost; simple operation; effective viscosity/pour-point reduction; minimal utility consumption relative to deep-conversion units.

Limitations: Low conversion (10–20 wt%); visbroken tar remains a fuel-oil-grade, low-value product and is thermally/oxidatively unstable (tendency to form sediment on storage/blending); products are highly olefinic and require downstream hydrotreating for use as transport fuel blend-stock; does not resolve the underlying strategic need to eliminate fuel-oil production.

4.1.2 Delayed Coking

Working Principle: Delayed coking is a semi-continuous, non-catalytic thermal cracking process that achieves near-complete rejection of the least valuable material as solid petroleum coke, converting the remainder of the vacuum residue into naphtha, gas oil and light gases.

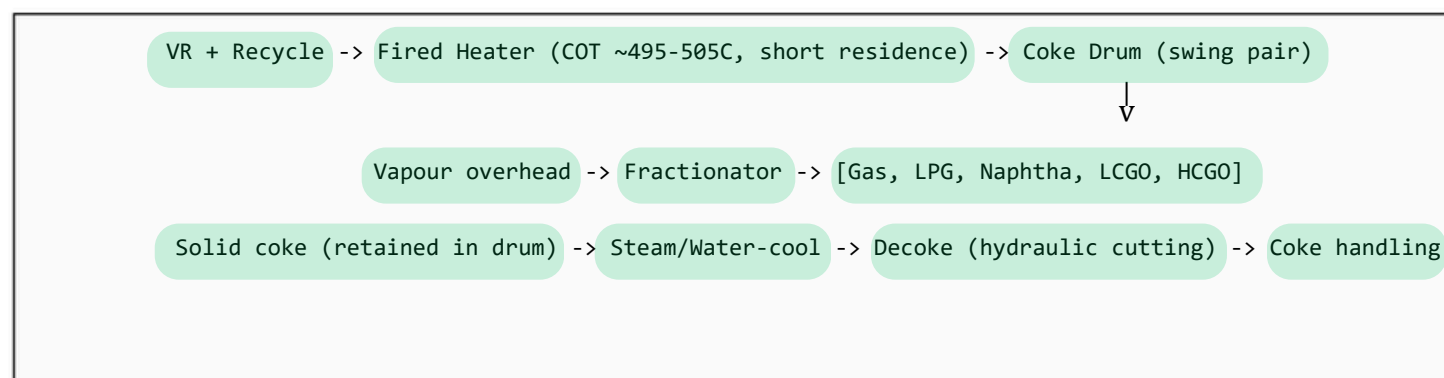
Process Description: Fresh VR feed (often combined with recycled heavy coker gas oil) is rapidly heated in a fired heater to approximately 495–505°C at the coil outlet, with a very short residence time in the heater tubes to avoid premature coking (fouling) inside the furnace tubes – cracking is deliberately “delayed” until the superheated feed enters one of a pair (or set) of large, insulated coke drums, where it is held for the 18–24- hour coking cycle. Vapour products continuously disengage from the top of the on-stream drum to a fractionator, while liquid remaining in the drum undergoes progressive thermal cracking and polymerization to form a porous, carbon-rich solid – green petroleum coke. Drums are operated in a cyclic, alternating (“swing”) sequence: one drum fills and cokes while the other is steamed, cooled, de-headed and hydraulically decoked (using high-pressure water jets, a step known as “cokecutting”).

Table 7: Typical Yield Distribution – Delayed Coking of Vacuum Residue

Product	Typical Yield (wt% of VR feed)
Fuel gas (C1–C2, H ₂ S, NH ₃)	6 – 10
LPG (C3–C4)	3 – 6
Coker naphtha	10 – 16
Coker gas oil (LCGO + HCGO)	40 – 55
Green petroleum coke	25 – 35

Source: Representative yield ranges compiled from Gary, Handwerk & Kaiser and Meyers, Handbook of Petroleum Refining Processes.

flow chart 7: Delayed Coking Unit – Simplified Process Flow Diagram



Advantages: Very high liquid-plus-coke conversion (65–85 wt%, and effectively near-total conversion of VR since coke itself is a saleable, though low-value, product); flexible feedstock tolerance, including very heavy, high-CCR, high-metals residues that would foul catalytic units; well-proven, robust technology with a long industrial track record.

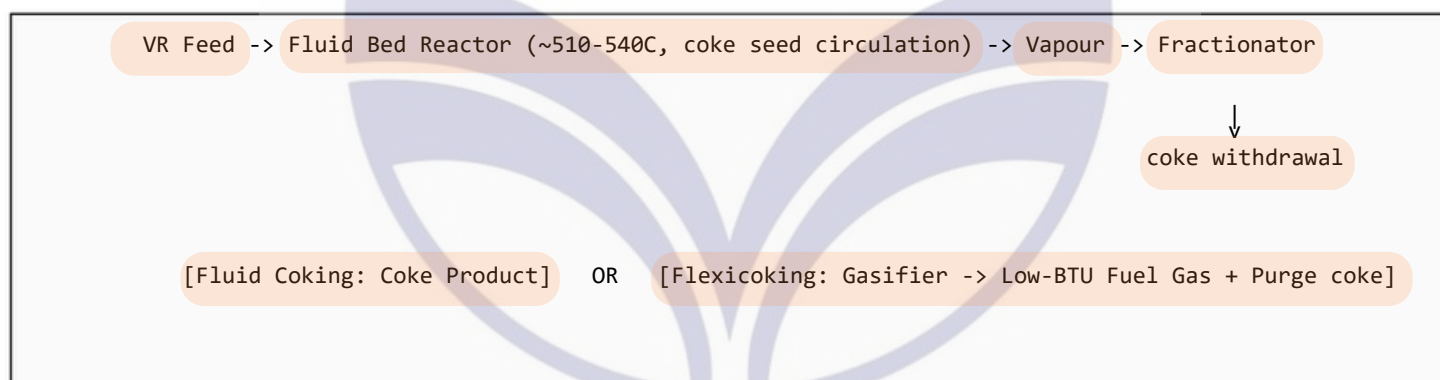
Limitations: Coke is a low-value, high-carbon by-product whose market value and environmental acceptability (particularly high-sulphur “fuel-grade” coke) are limited; product streams are highly olefinic and require substantial downstream hydrotreating capacity; cyclic operation and coke handling introduce mechanical complexity, dust/particulate handling requirements and periodic safety hazards during decoking; large land and capital footprint.

4.1.3 Fluid Coking and Flexicoking

Fluid coking replaces the batch coke-drum system with a continuous fluidized-bed reactor in which VR is sprayed onto a fluidized bed of hot coke particles maintained at 510–540°C; cracking occurs on the surface of circulating coke particles, which grow in size and are continuously withdrawn, while fresh “seed” coke is generated by attrition. Because reaction occurs at a bed surface rather than in bulk liquid, fluid coking achieves higher liquid yield and lower coke make than delayed coking for an equivalent feed.

Flexicoking is an extension of fluid coking in which the bulk of the coke produced is continuously gasified in an integrated gasifier vessel using air/oxygen and steam to produce a low-to-medium heating-value fuel gas (“coker gas”), which is used as refinery fuel after clean-up. This eliminates the marketing/disposal burden of solid petroleum coke almost entirely, at the expense of higher capital cost and the added complexity of gas clean-up (particularly removal of H₂S and particulate-bound metals).

Flow Chart 8: Fluid Coking / Flexicoking – Simplified Process Flow Diagram



Advantages (over delayed coking): Continuous operation (no cyclic drum switching); higher liquid yield; Flexicoking virtually eliminates solid coke disposal by converting it to usable fuel gas.

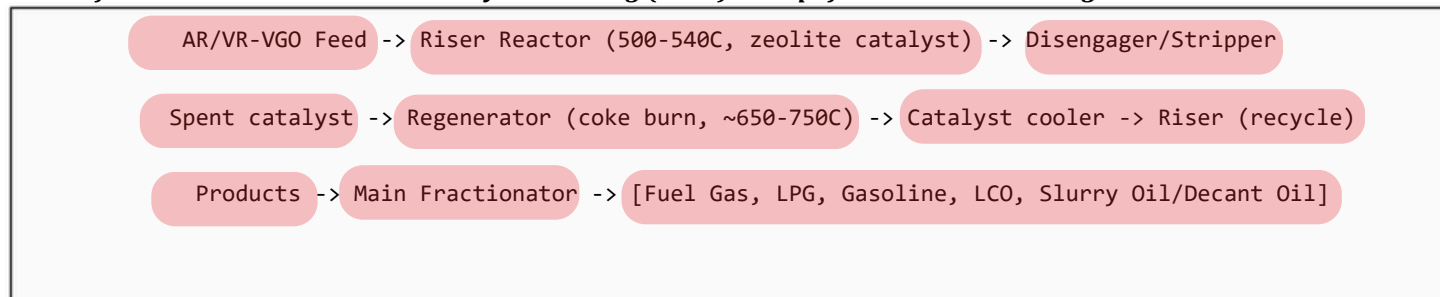
Limitations: Significantly higher capital and operating complexity; gasifier fuel gas requires clean-up of H₂S, particulates and trace metals before combustion; fewer licensors/reference units globally compared to delayed coking, limiting technology familiarity and spares availability.

4.2 Catalytic Processes – Residue Fluid Catalytic Cracking (RFCC)

Working Principle: RFCC extends conventional FCC technology, which normally processes vacuum gas oil, to accept atmospheric residue or residue-VGO blends as feed. Cracking proceeds through a carbocation (carbenium/carbonium ion) mechanism on the acid sites of a zeolitic (USY-type) catalyst at 500–540°C, converting large molecules predominantly into gasoline-range and LPG-range olefinic/paraffinic products, together with a coke deposit that is burned off the catalyst in a regenerator to supply the endothermic heat of reaction.

Because residue feeds carry substantially more CCR-forming material and metal contaminants (Ni, V) than VGO, RFCC units require significant hardware and catalyst modifications relative to standard FCC: a two-stage or high- efficiency regenerator to manage the higher coke burn and heat release; feed nozzles designed for improved atomization of the heavier feed; a catalyst cooler to remove excess regenerator heat; higher catalyst circulation and makeup rates to counteract metal poisoning and maintain activity; and metals-passivation additives (e.g., antimony- or bismuth based) mitigate the dehydrogenation activity of deposited of vanadium.

flow Chart 9: Residue Fluid Catalytic Cracking (RFCC) – Simplified Process Flow Diagram



Advantages: High-octane gasoline and olefin (propylene) yield valuable for petrochemical integration; moderate capital cost relative to hydrogen-addition routes; well-established technology base.

Limitations: Strict feed CCR and metals limits (feed pretreatment or blending with VGO is usually required); produces highly aromatic, high-sulphur cycle oil and slurry oil that require further hydrotreating; net hydrogen- deficient route (no external hydrogen addition), so cracked products are more olefinic/aromatic and less directly diesel-compatible than hydrocracked products; higher SO_x/particulate emissions from the regenerator, requiring flue-gas treatment.

4.3 Hydrogen-Addition (Resid Hydrocracking) Processes

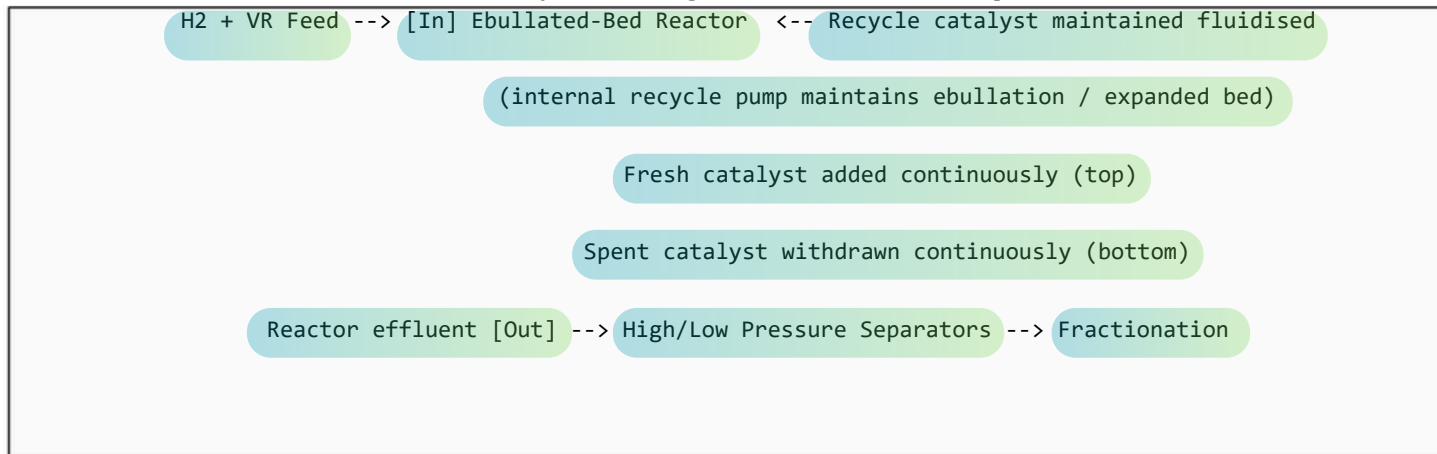
Hydrogen-addition processes are, in engineering and economic terms, the most effective route for deep, high-quality residue conversion. By supplying hydrogen under catalytic conditions (typically over a Mo/Ni or Mo/Co-promoted alumina-supported catalyst) at high pressure (100–200+ bar) and moderate-to-high temperature (400–450°C), cracked free-radical fragments are hydrogen-capped before they can polymerize into coke precursors, dramatically suppressing coke formation and allowing conversions of 60–95 wt% to be achieved as liquid product rather than solid coke.

4.3.1 Fixed-Bed and Ebullated-Bed Resid Hydrocracking

Fixed-bed resid hydroprocessing (e.g., conventional resid hydrodesulphurization/hydrocracking trains) uses a series of fixed catalyst beds and is well suited to lower-metals feeds, but catalyst deactivation by metal deposition and coke laydown progressively increases pressure drop and eventually forces a shutdown for catalyst change-out, limiting the feed metals content that can be processed economically.

Ebullated-bed (also called “ebullating” or fluidized liquid-phase) resid hydrocracking overcomes this limitation by suspending the catalyst as a fluidized bed within the up-flowing liquid/gas/catalyst mixture inside the reactor, using an internal recycle pump to maintain bed expansion (“ebullation”). This design permits continuous, online catalyst withdrawal and fresh-catalyst addition without shutting down the unit, allowing the process to tolerate very high feed CCR and metals content over long, continuous run lengths (commonly 3–5+ years between turnarounds).

Flow Chart 10: Ebullated-Bed Resid Hydrocracking – Generic Reactor Concept



4.3.2 LC-FINING™, H-Oil® and LC-MAX™ Technologies

LC-FINING™ (licensed by Chevron Lummus Global, CLG) and H-Oil® (originally developed by Cities Service/HRI, now also within the CLG/Axens technology family) are the two principal commercially proven ebullated-bed resid hydrocracking technologies deployed worldwide, both based on the same fundamental ebullated-bed reactor concept described above, differing mainly in proprietary internals, catalyst systems and reactor engineering detail. Typical VR conversion levels for LC-FINING™/H-Oil® installations range from about 60–80 wt%, producing naphtha, diesel/gas- oil range material and an unconverted oil (“tar” or resid pitch) that can be further processed by coking, gasification, or sold as low-sulphur fuel oil blend-stock/asphalt feedstock.

LC-MAX™ is CLG's latest-generation, higher-conversion evolution of the ebullated-bed platform, engineered to push VR conversion substantially higher (approaching ~90–95 wt%) than earlier-generation LC-FINING™ designs, while managing sediment stability and product quality through advanced reactor internals, staged reaction and integrated solvent-deasphalting/unconverted-oil management concepts. HPCL Visakh Refinery's Residue Upgradation Facility, commissioned in January 2026, is the world's first commercial LC-MAX™ installation, achieving approximately 93% conversion of bottom-of-the-barrel oils into high-value products – a conversion level substantially higher than typical first-generation ebullated-bed units.

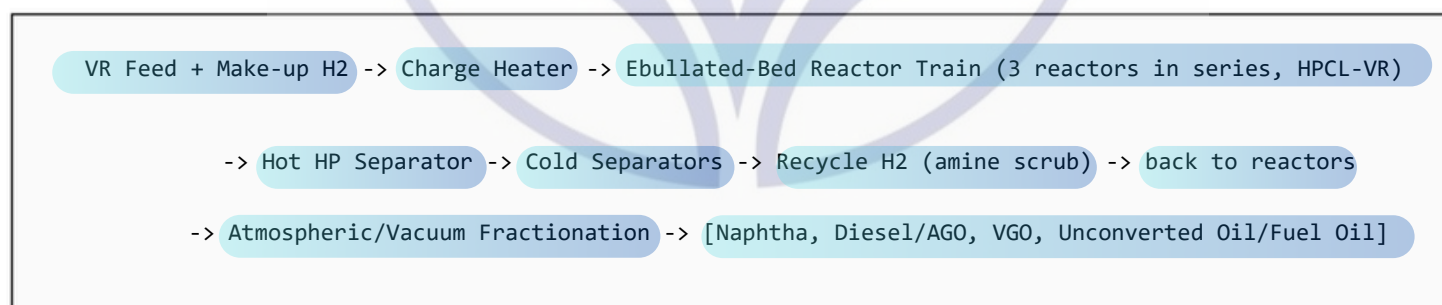


Table 9: Typical Operating Conditions – Ebullated-Bed Resid Hydrocracking (LC-MAX™-type)

Parameter	Typical Range
Reactor temperature	410 – 450°C
Reactor pressure (H ₂ partial pressure)	≈ 140 – 200 bar
Catalyst	Mo/Ni (or Mo/Co) on alumina extrudate, ebullated/fluidized
VR conversion (540°C+ basis)	60 – 80% (LC-FINING™/H-Oil®); up to ~93% (LC-MAX™, HPCL-VR)
Hydrogen consumption	≈ 150 – 300+ Nm ³ /m ³ of feed (grade-dependent)
HDS / HDM achieved (feed basis)	HDS 60–90%; HDM 60–95%

Source: Representative operating ranges compiled from CLG/Axens published technology descriptions and standard hydroprocessing literature; HPCL-specific conversion figure from HPCL press release, 05 January 2026.

Flow chart 11: LC-MAX™ / LC-FINING™ Ebullated-Bed Process Flow Diagram



Advantages: Very high liquid yield with minimal or no solid coke by-product; simultaneous deep hydrodesulphurization, hydrodenitrogenation and hydrodemetallization of the residue, substantially reducing downstream hydrotreating severity; tolerant of very heavy, high-CCR, high-metals feeds through continuous catalyst replacement; produces a product slate directly compatible with BS-VI diesel maximization strategy; long, continuous run lengths between turnarounds.

Limitations: Very high capital cost, driven by high-pressure metallurgy, large reactor vessels (HPCL's RUF reactors are approximately 2,200 MT each) and hydrogen-supply infrastructure; substantial hydrogen consumption requiring large-scale hydrogen generation capacity; catalyst handling (continuous withdrawal/addition of pyrophoric spent catalyst) requires stringent safety systems; sediment/asphaltene stability in the unconverted oil must be carefully managed to avoid fouling in downstream equipment; requires highly skilled operating and maintenance personnel.

4.3.3 Emerging Hydrogen-Addition Technology – Slurry-Phase Hydrocracking

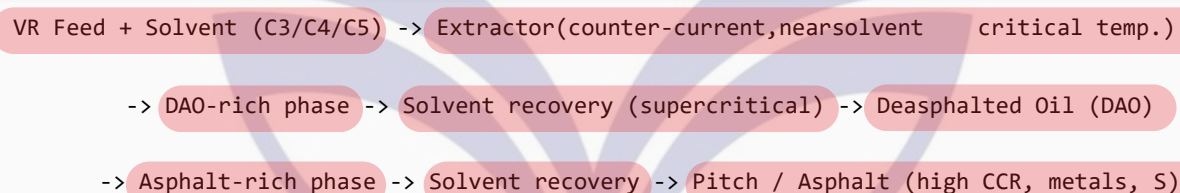
Slurry-phase (or “slurry-bed”) hydrocracking is a further evolution beyond ebullated-bed technology, in which an ultra-fine (nano- or micron-scale) dispersed catalyst or additive is suspended throughout the liquid phase rather than confined to a fluidized bed of extrudate particles. This provides intimate catalyst-to-feed contact and effectively eliminates diffusion limitations even for the largest asphaltenic molecules, enabling near-complete (>95%) conversion with strong resistance to sediment formation. Licensed technologies in this category (e.g., LC-SLURRY™, EST™, VCC™, Uniflex™) are at earlier stages of commercial deployment globally compared to ebullated-bed technology but represent the likely next frontier for refiners seeking near-total elimination of fuel-oil production.

4.4 Solvent-Based Processes – Solvent Deasphalting (SDA)

Working Principle: Solvent Deasphalting is fundamentally a physical liquid–liquid extraction process rather than a molecular conversion process. VR is contacted with a light paraffinic solvent (propane, butane, pentane, or blends) under moderate pressure at a temperature close to the solvent's critical point. The saturate- and lower-molecular- weight-resin fraction of the residue is highly soluble in the paraffinic solvent and is recovered as Deasphalted Oil (DAO); the asphaltene- and heavy-resin fraction, which is essentially insoluble in the paraffinic solvent, is rejected as a concentrated pitch (asphalt) stream carrying the bulk of the feed's metals, CCR and sulphur.

Because selectivity increases as the solvent becomes lighter (more paraffinic) and as extraction temperature approaches the solvent critical temperature, propane deasphalting yields the highest-quality (lowest CCR/metals), lowest-yield DAO, suited for lube-oil or FCC feedstock applications; butane and pentane deasphalting yield progressively higher DAO quantities of correspondingly lower quality, more suited as fuel-oil blend-stock or hydrocracker feed.

Flow chart 12: Solvent Deasphalting Unit – Simplified Process Flow Diagram



Advantages: Low operating temperature (thermally mild, essentially no cracking or coke formation); effective concentration/rejection of metals, CCR and asphaltenes into a small pitch stream, upgrading the bulk of the residue (as DAO) to a quality acceptable as feed to RFCC, hydrocracking, or lube-base-oil manufacture; relatively low capital cost compared to hydrogen-addition routes.

Limitations: SDA does not destroy or convert molecules – it only separates them, so overall barrel conversion is limited by DAO yield (typically 50–80 wt% depending on solvent and target quality); the rejected pitch remains a low-value, difficult-to-market by-product (suitable mainly as very heavy fuel oil, paving-grade bitumen feedstock, or gasification/coking feed); solvent handling introduces flammability and high-pressure safety considerations; DAO still commonly requires downstream hydrotreating before use as a premium fuel component. SDA is therefore most often deployed as a residue pretreatment step ahead of RFCC or hydrocracking, rather than as a stand-alone bottoms-elimination technology.

4.5 Emerging and Integrated Residue Conversion Technologies

- Slurry-phase / ebullated-bed hybrid hydrocracking (LC-SLURRY™, EST™, VCC™, Uniflex™): pushing conversion beyond 95% with improved sediment/asphaltene stability control.

- Integrated SDA + resid hydrocracking configurations: using SDA pitch rejection ahead of the hydrocracker to reduce feed metals/CCR load and extend catalyst life, or routing SDA pitch to a downstream coker/gasifier for near-zero-residue operation.
- Advanced catalyst technologies: graded catalyst systems (demetallization-, transition- and desulphurization-grade catalysts loaded in series) to extend cycle length and improve metals tolerance in ebullated-bed and fixed-bed reactors.
- Digitally optimized resid hydrocracking: real-time process monitoring, predictive analytics based on proprietary thermodynamic/kinetic models, and AI-driven optimization of reactor severity, hydrogen distribution and catalyst replacement rate – exemplified by HPCL's RUF (LC-MAX™) Digital Suite, described by HPCL as an industry-first deployment among Indian refineries.
- Residue gasification (Integrated Gasification Combined Cycle, IGCC) and partial oxidation: converting the lowest-value pitch/coke streams into syngas for hydrogen production or power generation, supporting a near- zero-residual-fuel-oil refinery configuration.
- Sustainable/renewable co-processing: exploratory co-processing of bio-based or waste-plastic-derived pyrolysis oils within resid hydrocracking trains to support refinery decarbonization objectives.

4.6 Comparative Assessment of Residue Upgradation Routes



Table 8: Comparison of Catalytic and Hydrogen-Addition Residue Upgradation Routes

Criterion	RFCC	Ebullated-Bed Resid Hydrocracking (LC-MAX™ type)	Solvent Deasphalting (SDA)
Conversion basis	Catalytic cracking (carbon rejection)	Hydrogen addition (catalytic hydrocracking)	Physical separation (no molecular conversion)
Typical conversion	60 – 75% (feed-quality dependent)	60 – 93% (LC-MAX™ upper end)	50 – 80% DAO yield
Product hydrogen content	Lower (olefinic/aromatic)	High (saturated, low-sulphur)	Unchanged (physical split only)
Feed metals/CCR tolerance	Limited (pretreatment usually needed)	High (continuous catalyst replacement)	High (rejects metals into pitch)
Capital intensity	Moderate	Very high	Moderate
Hydrogen consumption	None (net H ₂ producer via LPG/dry gas)	Very high	None
Suitability for BS-VI diesel maximization	Moderate (gasoline/olefins focus)	Excellent (direct diesel/distillate route)	Indirect (feed pretreatment role)

Table 10: Comparative Conversion and Product Quality – SDA, RFCC, Coking, Resid Hydrocracking (Directional Summary)

Technology	Conversion (wt%)	Sulphur removal	Solid by-product	Relative CAPEX
Visbreaking	10 – 20	Nil (no HDS)	None (tar product)	Low
Delayed Coking	65 – 85	Nil (no HDS; S concentrates in coke/gas oil)	Petroleum coke (25–35 wt%)	High
RFCC	60 – 75	Nil (S redistributes to LCO/slurry)	None (coke burnt in regenerator)	Moderate–High
Solvent Deasphalting	50 – 80 (DAO yield)	Partial (S concentrates in pitch)	Pitch/asphalt	Moderate
Ebullated-Bed Resid Hydrocracking (LC-MAX™)	60 – 93	High (60–90% HDS)	Minimal / none	Very High

CHAPTER 5: REACTION CHEMISTRY OF RESIDUE UPGRADATION

5.1 Thermal Cracking Reactions

Thermal cracking of residue hydrocarbons proceeds through a free-radical chain mechanism. Initiation occurs by homolytic cleavage of a C–C bond in a large paraffinic or naphthenic side-chain, generating two free radicals. Propagation proceeds via intramolecular hydrogen transfer (β -scission), producing a smaller olefin and a new, shorter-chain radical, which itself continues to react. Termination occurs by radical-radical recombination or disproportionation. Because thermal cracking is a purely carbon-rejection mechanism (no external hydrogen source), the overall hydrogen balance is satisfied by simultaneous condensation/polymerization reactions larger, more $R-CH_2-CH_2-R' \rightarrow R-CH=CH_2 + R'H$ (β -scission, simplified) condensed polynuclear-aromatic among the most hydrogen-deficient (aromatic, resinous, asphaltenic) fragments, which build up progressively structures that eventually precipitate as solid coke:
 (Aromatic/Asphaltenic fragments) $\rightarrow$ Coke (polynuclear aromatic solid) + H_2 (net, via disproportionation)

5.2 Catalytic Cracking Reactions (RFCC)

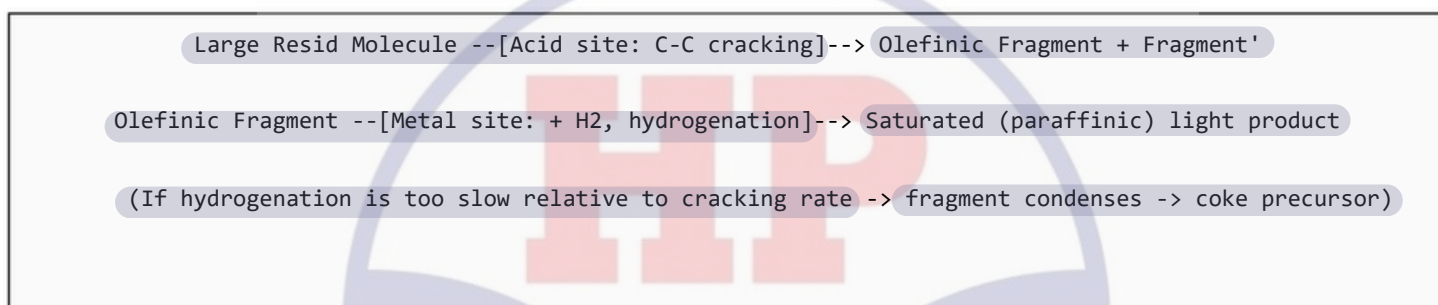
Catalytic cracking over acidic zeolite catalysts proceeds via a carbocation (carbenium/carbonium ion) mechanism, initiated by protonation of an olefin or by hydride abstraction from a paraffin/naphthene at a Brønsted or Lewis acid site. The resulting carbocation undergoes characteristic rearrangements – β -scission to give a smaller olefin and a new carbocation, hydride-shift isomerization, and hydrogen-transfer reactions between naphthenic

hydrogen-donor molecules and olefinic/aromatic hydrogen-acceptor molecules. Hydrogen-transfer reactions are particularly significant in RFCC of residues because they saturate a portion of the cracked olefins (improving gasoline stability) at the cost of dehydrogenating naphthenes/donor molecules towards aromatics and, ultimately, coke precursors – explaining why residue feeds with a higher aromatic/asphaltenic content produce disproportionately more catalyst coke than an equivalent VGO feed.

5.3 Catalytic Hydrocracking Reactions

Catalytic hydrocracking, as practiced in ebullated-bed resid hydrocracking (LC-MAX™/LC-FINING™/H-Oil® type units), is a bifunctional reaction system combining acidic cracking sites (on the alumina support) with metal hydrogenation sites (Mo/Ni or Mo/Co sulphides). Large residue molecules are first cracked at acid sites via a carbocation mechanism analogous to catalytic cracking; the resulting olefinic fragments are then rapidly hydrogenated at adjacent metal sites before they can undergo further condensation/oligomerization to coke precursors. This tightly coupled crack-then-hydrogenate sequence is the key mechanistic reason why hydrocracking achieves far higher liquid yield and far lower coke make than either pure thermal or pure catalytic (non-hydrogen) cracking of the same feed:

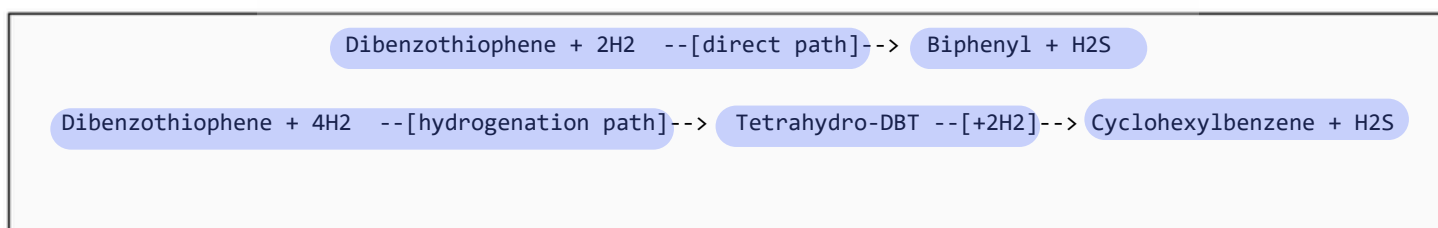
Flow Chart 13: Reaction Pathway – Catalytic Hydrocracking of Residue Molecules



5.4 Hydrodesulphurization (HDS)

Sulphur in residue is present mainly as thiophenes, benzothiophenes, dibenzothiophenes and their alkylated/condensed derivatives, together with sulphides and mercaptans. Over Mo/Ni or Mo/Co sulphide catalysts, HDS proceeds via two parallel pathways: direct desulphurization (hydrogenolysis of the C–S bond without prior ring saturation) and a hydrogenation pathway (partial aromatic-ring saturation followed by C–S hydrogenolysis), the formulation. relative contribution of each pathway depending on the specific sulphur-compound structure and catalyst

Flow chart 14: Hydrodesulphurization (HDS) Reaction Scheme



General HDS stoichiometry (illustrative): $R-S-R' + 2H_2 \rightarrow R-H + R'-H + H_2S$

5.5 Hydrodenitrogenation (HDN)

Nitrogen compounds in residue – pyridines, quinolines, carbazoles, indoles and their derivatives – are considerably more resistant to hydrotreatment than sulphur compounds and require prior aromatic-ring saturation before the C– N bond can be cleaved by hydrogenolysis, releasing ammonia. Basic nitrogen compounds are also strong competitive adsorbers on catalyst acid sites, and their incomplete removal is a major cause of downstream catalyst

General HDN stoichiometry (illustrative, quinoline): C_9H_6N (Quinoline) + $4H_2 \rightarrow$ Propylcyclohexylamine intermediate (e.g., hydrocracker, reformer) poisoning if residue-derived streams are insufficiently hydrotreated before further $\rightarrow$ (further hydrogenation/ring opening) $\rightarrow$ Propylbenzene + NH_3 processing.

5.6 Hydrodemetallization (HDM)

Nickel and vanadium in residue occur predominantly as metalloporphyrin complexes. HDM proceeds by hydrogenolytic cleavage of the metal–porphyrin ring system at catalyst pore-mouth sites, releasing the metal as a solid metal-sulphide deposit that remains irreversibly fixed on (and within the pores of) the catalyst, while the organic porphyrin fragment is further hydrocracked into lighter hydrocarbon products. Because HDM deposits accumulate physically on the catalyst rather than being carried out with the product, HDM catalysts are specifically engineered with large pore diameters and high metals-loading capacity (“guard-bed” or demetallization-grade catalyst), and in ebullated-bed units the continuous catalyst withdrawal/addition system exists primarily to manage

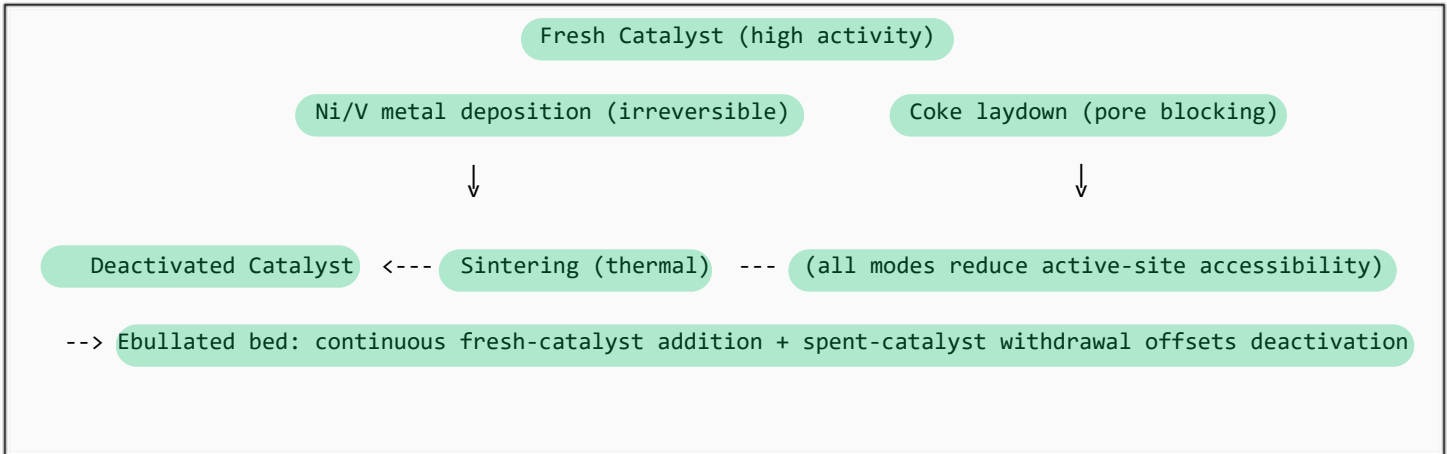
5.7 Coke Formation Reactions and Catalyst Deactivation Mechanisms

this metals-driven deactivation.

Coke formation is fundamentally a condensation-polymerization pathway competing with the desired hydrogenation/cracking reactions: asphaltenic and highly aromatic/resinous molecules, when insufficiently stabilized by hydrogen addition, undergo sequential dehydrogenation-condensation reactions to build progressively larger polynuclear-aromatic sheets, which ultimately precipitate as an insoluble, hydrogen-deficient solid (coke) either in the bulk liquid phase or on the catalyst surface.

- Coking on catalyst surface: pore-mouth and external-surface coke deposition blocks access of feed molecules to active sites, progressively reducing accessible surface area and activity (a diffusion-limited, feed-quality-dependent deactivation mode).
- Metal (Ni/V) poisoning: irreversible deposition of metal sulphides on active sites and within pore structures, permanently reducing catalyst activity and, for vanadium in particular, promoting parasitic dehydrogenation reactions that increase hydrogen consumption and gas make.
- Sintering/thermal deactivation: at excessive reactor temperature, active-metal crystallites can agglomerate (sinter), reducing dispersion and intrinsic activity; this is managed by maintaining reactor temperature within the catalyst's stable operating envelope and by the graded-catalyst-bed strategy.
- Sediment/asphaltene instability: incompletely stabilized asphaltenic fragments in the unconverted-oil product can precipitate downstream (in separators, heat exchangers, fractionator internals) if conversion severity exceeds the feed's inherent asphaltene-stability limit – a key design and operating constraint that is explicitly managed in advanced technologies such as LC-MAX™ through staged reaction, hydrogen distribution and, where required, integration with solvent-deasphalting of the unconverted oil.

Flow Chart 15: Catalyst Deactivation Mechanisms in Resid Hydroprocessing



CHAPTER 6: HPCL VISAKH REFINERY AND RESIDUE UPGRADATION

6.1 HPCL Visakh Refinery's Residue Upgradation Facility (RUF)

On 05 January 2026, HPCL officially announced the commissioning of the Residue Upgradation Facility (RUF) at its Visakh Refinery – described by the corporation as India's first Resid Hydrocracking Unit and the world's first commercial LC-MAX™ unit. The RUF is a hydrogen-based, bottom-of-the-barrel upgradation facility with a design capacity of 3.55 MMTPA, employing advanced LC-MAX™ ebullated-bed residue hydrocracking technology licensed by Chevron Lummus Global (CLG), with Larsen & Toubro (L&T) as the turnkey EPC contractor and Engineers India Limited (EIL) as project management consultant.



Table 11: HPCL Visakh Refinery – Residue Upgradation Facility (RUF) Design Data

Parameter	Value
Technology	LC-MAX™ Ebullated-Bed Resid Hydrocracking (world's first commercial unit)
Capacity	3.55 MMTPA
Licensors	Chevron Lummus Global (CLG)
EPC Contractor	Larsen & Toubro (L&T)
PMC	Engineers India Limited (EIL)
Conversion of bottom-of-barrel oils	≈ 93%
Reactor count / size	Three of the world's heaviest LC-MAX™ reactors, ≈ 2,200 MT each
Commissioning date	05 January 2026
Distillate yield uplift (vs pre-VRMP)	Up to 10 percentage points
Refinery Nelson Complexity Index (post-RUF)	11.6
Indigenization	Several critical equipment items manufactured indigenously, aligned with Aatmanirbhar Bharat

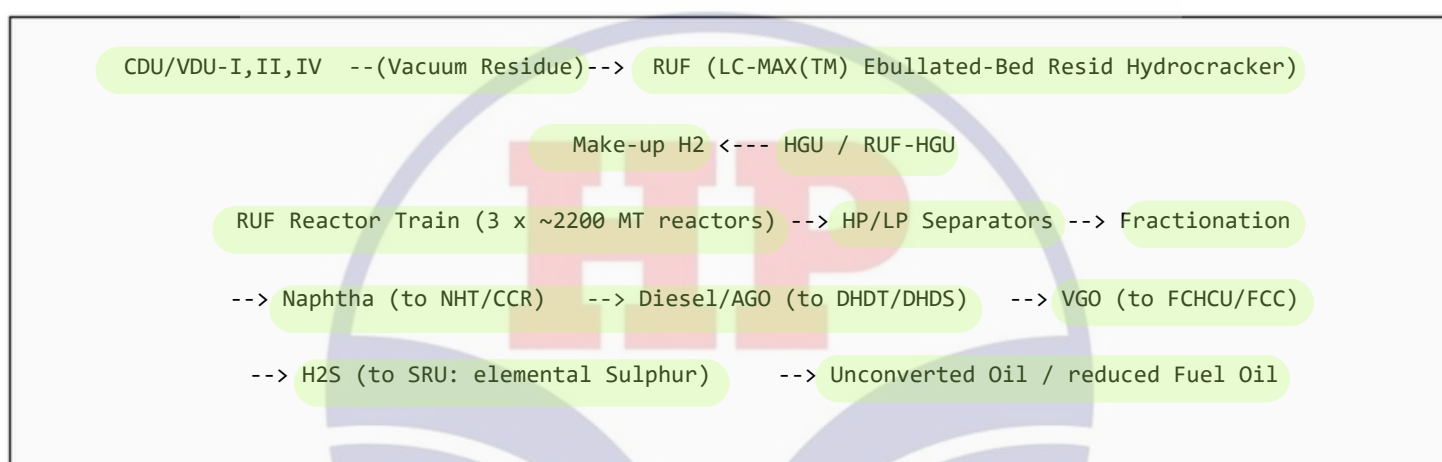
Source: HPCL official press release, "HPCL Commissions Residue Upgradation Facility with World's First LC-Max Unit at Visakh Refinery," 05 January 2026.

6.2 Process Description – RUF within the Visakh Refinery Configuration

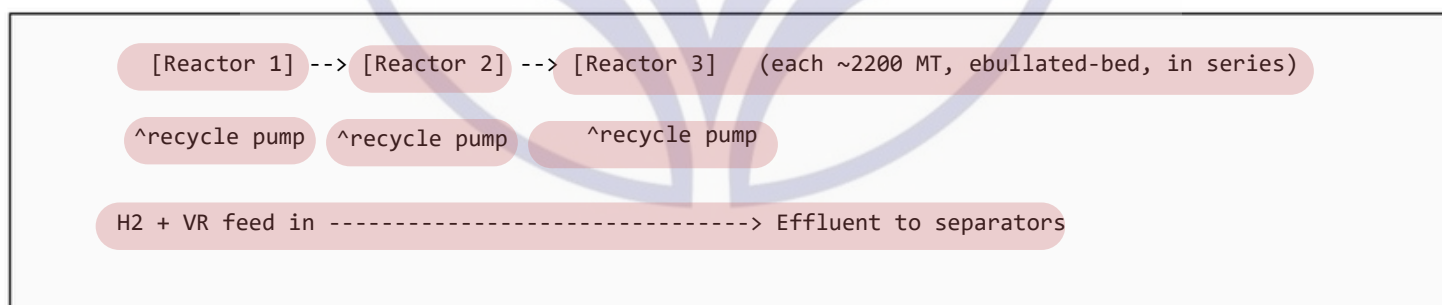
Within the post-VRMP refinery configuration, vacuum residue generated by the three parallel crude/vacuum distillation trains (CDU/VDU-I, II and the new 9 MMTPA CDU/VDU-IV) is routed to the RUF as feedstock, in place of being blended down into low-sulphur heavy stock (LSHS) or very-low-sulphur fuel oil (VLSFO) as was largely the case pre-RUF. Inside the LC-MAX™ reactor train, the vacuum residue is contacted with hydrogen-rich recycle and make-up gas (supplied by the refinery's Hydrogen Generation Units, HGU and RUF-dedicated HGU) over an ebullated bed of Mo/Ni-type catalyst maintained in continuous fluidized motion by internal recycle pumps, achieving simultaneous cracking, hydrogenation, hydrosulphurization, hydrodenitrogenation and hydrodemetallization of the residue.

Reactor effluent is separated in high- and low-pressure separator trains, with unreacted hydrogen scrubbed of H₂S (via amine absorption) and recycled to the reactors, while the liquid product is fractionated into naphtha, diesel/gas- oil range material, resid-derived VGO (routed to the FCHCU and/or FCC units for further conversion) and a much- reduced unconverted-oil/fuel-oil stream. Recovered H₂S from both the RUF and refinery-wide hydroprocessing units is processed in the Sulphur Recovery Units (SRU, including the dedicated VRMP SRU trains) to recover elemental sulphur, a saleable industrial product.

flow Chart 16: HPCL Visakh Refinery – Residue Upgradation Facility (RUF) Block Diagram



Flow Chart 17: LC-MAX™ Reactor Train at HPCL Visakh Refinery – Conceptual Layout



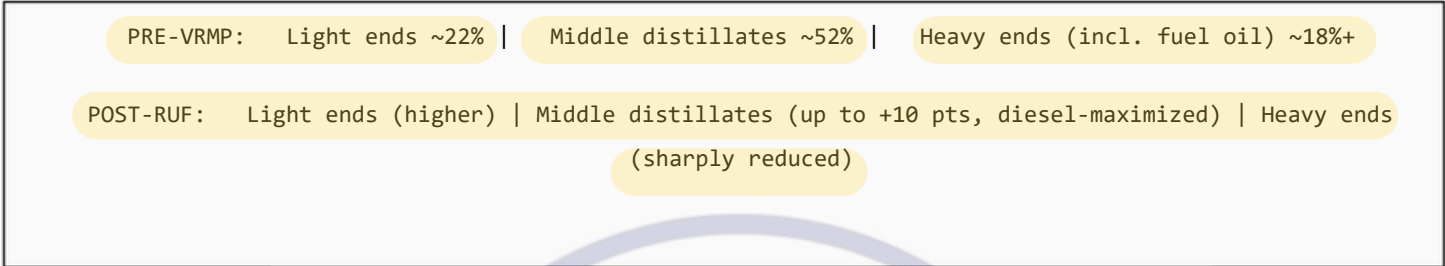
6.3 Bottoms-Upgradation Philosophy and Diesel Maximization

HPCL has articulated an explicit “diesel maximization” strategy for the Visakh Refinery, in which the RUF, together with the FCHCU commissioned in 2023, forms the core of an integrated bottoms-upgradation philosophy: crude residue that would previously have exited the refinery as discounted fuel oil is instead routed through progressively deeper conversion – first through the FCHCU (hydrocracking of VGO-range material to diesel and naphtha) and now through the RUF (hydrocracking of the vacuum-residue itself) – to maximize the yield of BS-VI- grade high-speed diesel and other high-value distillates per barrel of crude processed. This directly reduces the historical gap between HPCL's HSD marketing volumes and its own refining output, lowering dependence on external diesel sourcing and strengthening the corporation's supply chain.

6.4 Refinery Margin, Complexity and Digital Integration

HPCL has stated that the RUF is expected to substantially increase the Gross Refining Margin (GRM) of Visakh Refinery through a superior, distillate-rich product slate, the flexibility to process heavier and more economically discounted (“opportunity”) crude grades that would previously have been unattractive due to their high residue yield, and higher overall value realization per barrel of crude. Improved heat integration and process efficiency associated with the RUF have also been reported to reduce the refinery’s Energy Intensity Index (EII), lowering per- unit operating cost while advancing sustainability objectives. Complementing the physical plant, HPCL has deployed the RUF (LC-MAX™) Digital Suite – described as an industry-first deployment among Indian refineries – integrating real- time process monitoring, predictive analytics based on proprietary thermodynamic models, and AI-driven optimization to maintain stable, efficient and optimized ebullated-bed operation.

Flow Chart 18: Product Slate Shift – Pre-VRMP vs Post-RUF Visakh Refinery (Directional)



Source: Directional representation based on HPCL product-slate disclosures and VRMP/RUF distillate-yield-uplift statements; not exact refinery yield data.

6.5 Integration with Existing Process Units

The RUF does not operate in isolation; its output streams are fully integrated with the rest of the Visakh Refinery process configuration – naphtha to the NHT/CCR-Platformer/Isomerization block for BS-VI motor-spirit blending, diesel/gas-oil fractions to the DHDT/DHDS trains for final BS-VI HSD polishing, resid-derived VGO to the FCHCU and FCC units for further conversion to gasoline and petrochemical feedstock, off-gas and H2S to the Hydrogen Generation and Sulphur Recovery Units, and any residual unconverted oil to fuel-oil/VLSFO blending or, where economically favourable, further thermal/coking treatment. This full integration is what allows the RUF to lift refinery-wide distillate yield by up to 10 percentage points relative to the pre-VRMP baseline, rather than simply relocating the conversion burden elsewhere in the refinery.

6.6 Strategic Significance

The commissioning of the RUF represents a milestone not only for HPCL but for the Indian refining sector as a whole – it is described by HPCL as India's first Resid Hydrocracking Unit and, at a global level, the first commercial deployment of CLG's LC-MAX™ technology. The project also carries strategic significance under the Government of India's Aatmanirbhar Bharat (self-reliant India) initiative, with several critical high-pressure reactor and project equipment items manufactured indigenously – including reactors ranking among the heaviest LC-MAX™ vessels built anywhere in the world.

CHAPTER 7: PROCESS CALCULATIONS

This chapter presents representative sample calculations of the type used in evaluating residue properties and residue upgradation unit performance. All numerical inputs are illustrative/assumed for demonstration of method, unless explicitly referenced to Chapter 6 design data.

7.1 API Gravity and Density Conversion

API gravity is related to specific gravity (SG at 60°F/60°F) by the standard API formula:

$$API^{\circ} = (141.5 / SG) - 131.5$$

$$3): API^{\circ} = (141.5 / 1.02) - 131.5 = 138.7 - 131.5 = 7.2^{\circ} API$$

Sample calculation: For a vacuum residue of specific gravity SG = 1.02 (representative HPCL-VR feed range, Table

This places the sample within the “heavy residue” API range of 2–15° identified in Table 3, consistent with typical

7.2 Conradson Carbon Residue (CCR) – Blending Estimate

vacuum residue derived from the medium-to-high-sulphur crude slate (Arab Light/Heavy, Basrah Medium, Upper Zakum, Urals, etc.) processed at Visakh Refinery. CCR of a blended feed is commonly estimated (for engineering screening purposes) by a linear weight-fraction blending rule: $CCR(\text{blend}) = \sum [w_i \times CCR_i]$

Sample calculation: A resid hydrocracker feed comprising 70 wt% vacuum residue (CCR = 22 wt%) blended with 30 wt% heavy vacuum gas oil recycle (CCR = 2 wt%):

Table 12: Sample CCR and API Gravity Calculation Data Set

Stream	wt% of Blend	CCR (wt%)	API Gravity (°)
Vacuum Residue (VR)	70	22.0	7.2
HVGO recycle	30	2.0	18.5
Blended RUF Feed	100	16.0	–

7.3 Conversion Calculation

Residue conversion is conventionally defined on a cut-point basis (commonly 540°C or 565°C TBP-equivalent):

Sample calculation: For an LC-MAX™-type unit processing 100 t/h of VR feed (100% 540°C+ by definition) yielding

7 t/h of unconverted 540°C+ oil: This matches HPCL's reported RUF conversion of approximately 93% of bottom-of-the-barrel oils into higher-value products, consistent with Table 11.

7.4 Sulphur (Mass) Balance and HDS Estimation

sulphur-mass basis:

Sulphur removed (as H₂S) is calculated from a simple mass balance across the reactor system:

Sample calculation: Feed rate 10,000 t/d VR at 4.5 wt% S; liquid product yield 9,300 t/d at 0.6 wt% S (HDS ~90% as per Table 9 upper range):

$$\text{Sulphur in product} = 9,300 \times 0.006 = 55.8 \text{ t/d}$$

$$\text{Sulphur removed (as H}_2\text{S, to SRU)} \approx 450 - 55.8 = 394.2 \text{ t/d}$$

$$\text{HDS (\%)} = (394.2 / 450) \times 100 \approx 87.6\%$$

7.5 Hydrogen Consumption Estimation

Hydrogen consumption in resid hydrocracking is driven by three concurrent mechanisms: (i) hydrogenation of cracked olefinic fragments, (ii) heteroatom removal (HDS, HDN, HDM, each consuming stoichiometric hydrogen), and (iii) aromatic ring saturation. A simplified engineering estimate expresses total consumption as a function of conversion and feed quality: $\text{H}_2 \text{ consumption (Nm}^3\text{/m}^3 \text{ feed)} \approx k \times \text{Conversion(wt\%)} \times [\text{CCR(feed) factor}] \times [\text{S(feed) factor}]$

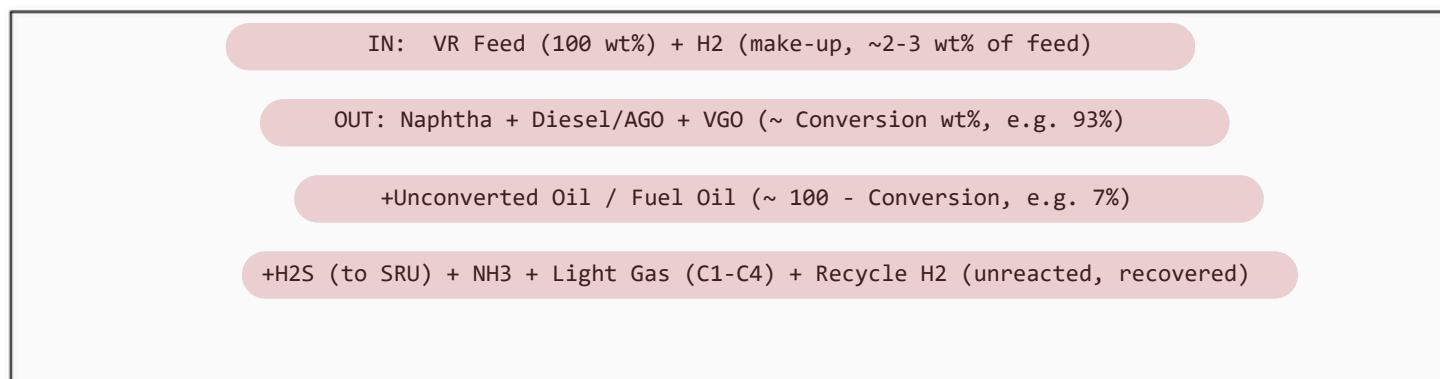
Table 13: Sample Hydrogen Consumption Estimate for Resid Hydrocracking

Basis	Value
Feed rate	10,000 t/d ($\approx$ 3.55 MMTPA basis, RUF scale)
Conversion achieved	93%
Typical specific H ₂ consumption (Table 9)	$\approx$ 250 Nm ³ /m ³ of feed (mid-range estimate)
Approx. feed volume (SG $\approx$ 1.0)	$\approx$ 10,000 m ³ /d
Estimated H ₂ consumption	$\approx$ 2.5 million Nm ³ /d (order-of-magnitude estimate)

Such estimates illustrate why large-scale resid hydrocracking installations, including HPCL's RUF, require correspondingly large dedicated Hydrogen Generation Unit capacity (HGU/RUF-HGU) to sustain make-up hydrogen supply.

7.6 Simplified Mass Balance Envelope

Flow chart 19: Mass Balance Envelope for a Typical Resid Hydrocracking Unit



7.7 Refinery Economics – Illustrative Margin Uplift Calculation

A simplified illustrative margin-uplift calculation (not HPCL-specific financial data) demonstrates the economic logic of residue upgradation: assume VR realizes a notional value of crude price minus US\$15/bbl, while the

Uplift = 900 bbl/d × [8 - (-15)] = 900 × 23 = US\$20,700/d (illustrative only)

blended distillate slate produced from that VR (diesel, naphtha, VGO) realizes crude price plus US\$8/bbl.

Converting 1,000 bbl/d of VR at 90% conversion to distillate therefore creates a margin uplift of approximately:

This type of calculation – scaled to actual crude differentials, product cracks and unit throughput – underlies

CHAPTER 8: ENVIRONMENTAL AND SUSTAINABILITY ASPECTS

HPCL's stated expectation of substantially improved Gross Refining Margin following RUF commissioning.

8.1 Carbon Footprint Reduction

By converting vacuum residue – the most carbon-intensive, hydrogen-deficient fraction of the barrel – into hydrogen- rich distillate products, resid hydrocracking technologies such as LC-MAX™ reduce the overall carbon intensity of the refinery's product slate relative to a configuration that blends the same residue into high-carbon fuel oil. HPCL has reported that improved heat integration and process efficiency associated with the RUF have reduced the refinery's Energy Intensity Index (EII), directly lowering the specific energy consumption, and hence the specific CO2 emissions, per tonne of product.

8.2 Sulphur Emission Reduction

Deep hydrodesulphurization within the RUF, combined with the refinery's existing and VRMP-augmented Sulphur Recovery Units (SRU), converts organic sulphur that would otherwise be emitted as SOx (if burnt as high-sulphur fuel oil) or remain in low-value fuel-oil products, into elemental sulphur – a saleable industrial commodity – while producing BS-VI-compliant, ultra-low-sulphur (10 ppm) motor fuels. HPCL has stated its objective to maintain refinery-wide SOx emissions at or below the existing permissible limit of 11.5 tonnes per day despite the substantial capacity and complexity increase delivered by VRMP and RUF.

8.3 Cleaner Fuel Production and Product Quality

The RUF directly supports the refinery's transition to a fully BS-VI-compliant product slate by supplying hydrotreated, low-sulphur, low-nitrogen distillate feedstocks to the downstream DHDT/DHDS trains, reducing the

hydrotreating severity (and hence energy/hydrogen consumption) required at those units to reach final product specification.

8.4 Energy Efficiency and Heat Integration

Modern resid hydrocracking units employ extensive feed/effluent heat exchange, pinch-optimized heat-recovery networks, and waste-heat recovery from hot separator and fractionator streams to minimize fired-heater duty. Such heat integration measures are central to the Energy Intensity Index improvement HPCL has associated with the RUF.

8.5 Waste Minimization and Effluent Treatment

Residue upgradation technologies vary considerably in their solid/liquid waste profile: thermal coking processes generate a substantial solid coke by-product requiring dedicated handling, storage and marketing infrastructure, whereas hydrogen-addition technologies such as LC-MAX™ generate minimal solid waste, with spent hydroprocessing catalyst (containing recoverable molybdenum, nickel and vanadium values) as the principal solid residual, typically sent for off-site metals reclamation. Visakh Refinery's broader effluent management system – comprising primary, secondary and tertiary treatment stages, oil-recovery systems and ETP sludge management – is designed to handle the aqueous effluent (sour water, wash water) generated across all refinery process units, including the RUF.

8.6 Hydrogen Utilization and the Hydrogen Economy

Because resid hydrocracking is inherently hydrogen-intensive, the RUF is tightly integrated with the refinery's Hydrogen Generation Units. Efficient hydrogen utilization – through high-pressure recycle-gas compression and amine-based H₂S scrubbing to maximize hydrogen recycle purity – is essential both for economic operation and for minimizing the refinery's overall natural-gas/feedstock consumption associated with hydrogen manufacture, a theme of increasing relevance as refiners plan for eventual integration of low-carbon (“blue” or “green”) hydrogen supply.

8.7 Sustainability Initiatives

- ISO 14001 environmental management system implementation across Visakh Refinery.
- Continuous ambient air quality monitoring with connectivity to APPCB/CPCB regulatory monitoring systems.
- Leak Detection and Repair (LDAR) programme for fugitive VOC emissions.
- Low-NO_x burner technology in furnaces/heaters and use of low-sulphur refinery fuel gas and naphtha.
- Recirculating cooling-water systems and maximized recycle/reuse of treated effluent, targeted at minimizing fresh-water withdrawal.
- Digital optimization (RUF LC-MAX™ Digital Suite) to maintain stable, efficient operation and reduce off-spec/upset-related emissions and waste.

Table 14: Environmental Performance Indicators – Pre- and Post-RUF Visakh Refinery (Directional)

Indicator	Pre-RUF / Pre-VRMP	Post-RUF (Stated Objective / Reported Direction)
SOx emissions	Baseline (≤ 11.5 TPD ceiling)	Maintained at ≤ 11.5 TPD despite capacity increase
Energy Intensity Index (EII)	Baseline	Reduced, per HPCL disclosure
Distillate yield (wt% on crude)	≈ 76 wt% (pre-VRMP target basis)	≈ 86 wt% (VRMP target); further uplift up to 10 pts from RUF
Fuel oil / VLSFO make	Higher	Substantially reduced
Refinery complexity (NCI)	Lower	11.6

CHAPTER 9: INDUSTRIAL SAFETY IN RESIDUE UPGRADATION OPERATIONS

9.1 General Refinery Safety Philosophy

HPCL Visakh Refinery operates under a structured, multi-layered safety management philosophy encompassing engineered safeguards (fire-resistant materials, firewalls, blast-resistant control rooms), procedural safeguards (permit-to-work, lock-out/tag-out, management of change), and organizational safeguards (regular safety training, mockdrills, and a dedicated emergency response organization with trained firefighting teams and access to foam and water-cannon systems).

9.2 High-Pressure Equipment Safety

Residhydrocracking units such as the RUF operate at very high hydrogen partial pressures (of the order of 140–200 bar) and elevated temperature, and therefore demand stringent mechanical-integrity management: hydrogen- service metallurgy selected to resist hydrogen embrittlement and high-temperature hydrogen attack (per API RP 941 Nelson curves), rigorous pressure-vessel inspection and non-destructive testing regimes, relief and blowdown systems sized for credible overpressure scenarios, and high-integrity instrumented protective systems (SIS/SIL- rated trip systems) on critical reactor and separator loops.

9.3 Hydrogen Handling Safety

Hydrogen's wide flammability range, low ignition energy and propensity to form invisible flames make leak detection and ignition-source control especially critical around the RUF, associated Hydrogen Generation Units, and interconnecting high-pressure piping. Key safeguards include continuous hydrogen/hydrocarbon gas detection networks, area classification and intrinsically safe electrical design in hydrogen-handling zones, nitrogen purging protocols for equipment opening/entry, and strict control of hot-work permits in hydrogen-service areas.

9.4 Residue Processing Hazards

- Pyrophoric spent catalyst: catalyst continuously withdrawn from the ebullated-bed reactors can be pyrophoric on exposure to air and requires controlled cooling, inerting and wetting/passivation procedures during handling and disposal.
- High-temperature residue and coke handling: contact with hot residue, coke or catalyst presents severe burn hazards; delayed-coking-type units additionally carry specific hazards during drum de-heading and hydraulic decoking (high-pressure water-jet operations).
- H₂S exposure: sour gas and sour water streams throughout the residue-upgradation and sulphur-recovery train require personal and area H₂S monitoring, forced-ventilation design, and emergency escape-breathing-apparatus availability.
- Asphaltene/sediment fouling: uncontrolled sediment formation in unconverted-oil handling systems can create localized hot-spots and blockages, elevating fire risk if not managed through the process design safeguards discussed in Chapter 5.

9.5 Fire and Explosion Hazards

Given the high hydrocarbon and hydrogen inventories, elevated operating temperatures and pressures, and flammable vapour potential throughout the residue-upgradation complex, fire and explosion risk management follows a defence-in-depth approach: inherently safer design (minimizing inventory and simplifying high-hazard interconnections), passive protection (fireproofing of structural steel and critical instrument/control cabling), active protection (fixed and mobile firewater/foam systems, deluge systems on high-hazard vessels), and procedural controls (area classification, hot-work permitting, and emergency shutdown philosophy).

9.6 HPCL Visakh Refinery Safety Practices – Summary

- Advanced fire detection systems, fire-resistant construction and regular inspection regimes across the refinery, including the RUF complex.
- Well-defined emergency response plan with trained firefighting teams, evacuation procedures and firefighting equipment including foam and water cannons.
- Regular, structured safety training for personnel to build hazard awareness and emergency preparedness.
- Environmental protection measures integrated with fire-safety planning to prevent spills and limit ecological impact in the event of an incident.
- Full compliance with applicable statutory safety and environmental regulations.
- Continuous review and improvement of safety protocols against evolving industry best practice.
- Proactive community engagement and public awareness of emergency procedures around the refinery.

CHAPTER 10: FUTURE SCOPE

10.1 Next-Generation Residue Processing Technologies

Slurry-phase hydrocracking technologies, offering conversion levels beyond 95% with strong resistance to sediment/asphaltene instability, represent the likely next step-change in residue upgradation beyond ebullated-bed technology, and are of direct relevance to refiners seeking to approach a “near-zero-fuel-oil” refinery configuration. Continued evolution of ebullated-bed technology itself – as exemplified by the progression from LC-FINING™/H-Oil® to LC-MAX™ – is also expected to continue, with further gains in conversion, sediment stability and catalyst life.

10.2 Artificial Intelligence and Digitalization in Refineries

The deployment of the RUF (LC-MAX™) Digital Suite at Visakh Refinery – integrating real-time monitoring, predictive analytics based on proprietary thermodynamic models and AI-driven optimization – is illustrative of a broader industry trend towards digital twins, advanced process control and machine-learning-based predictive maintenance in complex hydroprocessing units. Future development is expected to extend such digital capability

towards refinery-wide, closed-loop optimization spanning crude selection, unit-level severity optimization and product-blending, with residue-upgradation units – given their complexity and sensitivity to feed variability – likely to remain at the forefront of such digital adoption.

10.3 Net-Zero and Low-Carbon Refining Concepts

As refiners globally develop pathways towards net-zero operations, residue-upgradation strategy is expected to increasingly intersect with carbon-management strategy: integration of carbon capture on hydrogen-generation and fired-heater flue gas streams, transition of make-up hydrogen supply towards lower-carbon-intensity (“blue”, and eventually “green”, electrolytic) hydrogen, and exploration of residue gasification routes that can simultaneously eliminate fuel-oil production and generate low-carbon hydrogen or power.

10.4 Green Refining and Circular Feedstocks

Co-processing of renewable and circular feedstocks – such as bio-oils derived from pyrolysis of biomass or waste plastics – within existing resid hydrocracking or hydrotreating trains is an active area of technology development globally, offering refiners a pathway to incrementally reduce the fossil-carbon intensity of their product slate using existing high-pressure hydroprocessing assets.

10.5 Hydrogen Economy Implications

The scale of hydrogen demand created by deep residue conversion (Chapter 7) positions large resid-hydrocracking refiners as natural anchor consumers, and potential future producers/exporters, within an emerging regional hydrogen economy. As indigenous electrolytic hydrogen and blue-hydrogen production scale up in India under Rpepfilnye.ry post-RUF – are well positioned to integrate lower-carbon hydrogen supply into their operations over time.

10.6 Continued Capacity and Complexity Growth

national hydrogen-mission initiatives, refineries with substantial existing hydrogen infrastructure – such as Visakhu Having reached a Nelson Complexity Index of 11.6, HPCL Visakh Refinery is positioned among India's leading deep-conversion refineries; continued evolution of its configuration – whether through further debottlenecking of existing units, incremental capacity creep on the RUF and FCHCU, or future petrochemical integration projects – is a natural extension of the technology and investment trajectory documented in this report.

CHAPTER 11: CONCLUSION

Residue upgradation has evolved from a peripheral refinery activity concerned chiefly with viscosity and pour-point reduction of fuel oil, into one of the central strategic and technological pillars of modern petroleum refining. The structural decline of high-sulphur fuel-oil demand, the tightening of clean-fuel regulation (culminating, in the Indian context, in BS-VI norms), and the direct economic incentive to extract maximum distillate value from every barrel of crude have together made deep bottom-of-the-barrel conversion an operational necessity for globally competitive refiners.

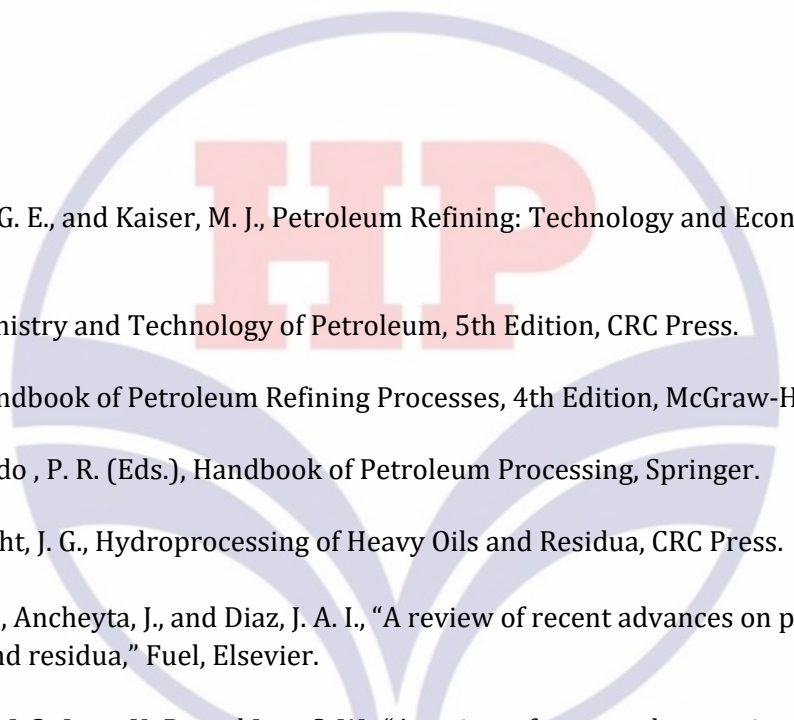
This report has surveyed the full spectrum of residue upgradation technologies – thermal processes (visbreaking, delayed coking, fluid coking and flexicoking), catalytic processes (Residue Fluid Catalytic Cracking), hydrogen- addition processes (fixed-bed and ebullated-bed resid hydrocracking, including LC-FINING™, H-Oil® and the latest- generation LC-MAX™ technology, as well as emerging slurry-phase hydrocracking), and solvent-based processes (Solvent Deasphalting) – examining, for each, the underlying process chemistry, reaction mechanisms, operating conditions, product yields, industrial advantages and limitations.

Among these routes, hydrogen-addition technology, and specifically ebullated-bed resid hydrocracking, emerges as the most technically and economically effective option for refiners pursuing simultaneous objectives of maximizing distillate yield, achieving deep sulphur/nitrogen/metals removal, and minimizing solid or heavy by-product

generation. HPCL Visakh Refinery's own Residue Upgradation Facility – commissioned in January 2026 as India's first Resid Hydrocracking Unit and the world's first commercial LC-MAX™ installation – exemplifies this technology choice at an unprecedented scale for the Indian refining industry, achieving approximately 93% conversion of bottom-of-the-barrel oils, lifting refinery distillate yield by up to 10 percentage points, raising the refinery's Nelson Complexity Index to 11.6, and materially strengthening its Gross Refining Margin and BS-VI clean-fuel production capability.

Beyond its immediate technical and economic significance, the RUF project illustrates the broader trajectory of the Indian refining sector: increasing refinery complexity and crude-flexibility, deepening integration of hydroprocessing across the barrel, growing reliance on digital process optimization, and a gradual but decisive shift away from residual fuel oil towards a distillate- and petrochemical-oriented product slate. As the industry looks towards an eventual low-carbon future, the hydrogen infrastructure, deep-conversion capability and digital optimization platforms established through projects such as HPCL's RUF are likely to serve as a foundation for the next generation of sustainable, integrated refining-petrochemical-hydrogen complexes in India.

REFERENCES

- 
- [1] Gary, J. H., Handwerk, G. E., and Kaiser, M. J., Petroleum Refining: Technology and Economics, 5th Edition, CRC Press, Taylor & Francis.
- [2] Speight, J. G., The Chemistry and Technology of Petroleum, 5th Edition, CRC Press.
- [3] Meyers, R. A. (Ed.), Handbook of Petroleum Refining Processes, 4th Edition, McGraw-Hill.
- [4] Jones, D. S. J., and Pujado, P. R. (Eds.), Handbook of Petroleum Processing, Springer.
- [5] Ancheyta, J., and Speight, J. G., Hydroprocessing of Heavy Oils and Residua, CRC Press.
- [6] Rana, M. S., Samano, V., Ancheyta, J., and Diaz, J. A. I., "A review of recent advances on process technologies for upgrading of heavy oils and residua," Fuel, Elsevier.
- [7] Sahu, R., Song, B. J., Im, J. S., Jeon, Y.-P., and Lee, C. W., "A review of recent advances in catalytic hydrocracking of heavy residues," Journal of Industrial and Engineering Chemistry.
- [8] Hindustan Petroleum Corporation Limited, "HPCL Commissions Residue Upgradation Facility with World's First LC-Max Unit at Visakh Refinery, Advancing Deep-Conversion Refining Capability and Enhancing Profitability," Press Release, Mumbai, 05 January 2026.
- [9] HPCL Visakh Refinery, Corporate Presentation and Plant Orientation Notes (Utilities, Environmental Management, Process Units, VRMP, Crude Slate and Product Profile), HPCL Visakh Refinery, Visakhapatnam.
- [10] Hindustan Petroleum Corporation Limited, Annual Report (latest available edition), HPCL, Mumbai.
- [11] Ministry of Petroleum and Natural Gas, Government of India, Petroleum Planning and Analysis Cell (PPAC) reports and statistics.
- [12] International Maritime Organization, "IMO 2020 – Cutting Sulphur Oxide Emissions," IMO, London.
- [13] Chevron Lummus Global (CLG), LC-FINING™ and LC-MAX™ Technology Descriptions, published technology literature.

[14] Axens, H-Oil® Ebullated-Bed Residue Hydrocracking Technology, published technology literature.

[16] Bureau of Indian Standards / Ministry of Road Transport and Highways, Bharat Stage VI (BS-VI) Emission Norms and Fuel Specifications, Government of India.

[17] Geddam, K., Akash, B., Devi, P. Y., et al., "Industrial Internship Report on A Study on Crude Distillation Unit," HPCL Visakh Refinery, 2024 (referenced for institutional report formatting and refinery background only).

